



TECHNICAL TECHNICAL TECHNICAL
REGULATIONS REGULATIONS REGULATIONS



Contents

Help text

Article T1 Definitions

- T1.1 STEM Racing™ India car
- T1.2 Fully assembled car
- T1.3 Body
- T1.4 Power unit cartridge chamber
- T1.5 Wing
- T1.6 Wing support structure and end plate
- T1.7 Nose cone
- T1.8 Wheel
- T1.9 Wheel support system
- T1.10 Tether line guide
- T1.11 Surface finish and decals
- T1.12 STEM Racing™ India logo decal
- T1.13 Hand finishing
- T1.14 Official STEM RACING™ Model Block
- T1.15 Engineering drawings
- T1.16 Renderings
- T1.17 Reference planes
- T1.18 Additional components
- T1.19 Normal
- T1.20 Full 8-gram power unit cartridge
- T1.21 Halo
- T1.22 Legal ballast

Article T2 General principles

- T2.1 Regulations documents
- T2.2 Interpretation of the regulations
- T2.3 Amendments to the regulations
- T2.4 Classification of regulations
- T2.5 Compliance with regulations
- T2.6 Design ideas and regulation compliance questions
- T2.7 Measurements
- T2.8 Benefit of doubt
- T2.9 Spirit of the competition
- T2.10 Originality of work

Article T3 Fully assembled car

- T3.1 Design and manufacture
- T3.2 Safe construction
- T3.3 Defined features
- T3.4 Total width
- T3.5 Total height
- T3.6 Total weight
- T3.7 Track clearance
- T3.8 Status during racing
- T3.9 Replacement components

Article T4 Body

- T4.1 Body construction

T4.2	Virtual cargo
T4.3	Virtual cargo identification
T4.4	Halo
T4.5	Helmet
T4.6	STEM Racing™ India logo decal location
T4.7	Team number
T4.8	Decal thickness

Article T5 Power unit cartridge chamber

T5.1	Diameter
T5.2	Distance from track surface
T5.3	Depth
T5.4	Max angle of chamber
T5.5	Chamber safety zone
T5.6	Power unit cartridge visibility

Article T6 Tether line guides

T6.1	Location
T6.2	Internal dimension
T6.3	Tether line guide safety

Article T7 Wheels and wheel support structures

T7.1	Number and location
T7.2	Distance between opposing wheels
T7.3	Wheelbase
T7.4	Track contact width
T7.5	Diameter
T7.6	Race track contact
T7.7	Rolling surface
T7.8	Rotation
T7.9	Visibility in top and bottom views
T7.10	Visibility in side views
T7.11	Visibility in front view
T7.12	Wheel support
T7.13	Wheel safety test

Article T8 Nose, front wing, end plates and wing support structures

T8.1	Nose, front wing, end plates and wing support structure identification
T8.2	Nose cone assembly dimension
T8.3	Front wing(s) description and placement
T8.4	Front wing(s) construction and rigidity
T8.5	Nose, front wing(s), front wing end plates and wing support location
T8.6	Front wing dimensions
T8.7	Front wing clear airflow
T8.8	Front wing visibility

Article T9 Rear wing and wing support structures

T9.1	Rear wing and wing support structure identification
T9.2	Rear wing description and placement
T9.3	Rear wing construction and rigidity
T9.4	Rear wing, wing support structure and rear overhang location
T9.5	Rear wing dimensions
T9.6	Rear wing clear airflow
T9.7	Rear wing visibility

Article T10 Additional components

T10.1 Description and placement

Appendix i Start box and finish gate

Appendix ii Official STEM Racing™ Model Block dimensions

Appendix iii Front wing compliance examples

Appendix iv Rear wing compliance examples

Appendix v Wing overlap

Appendix vi Official Denford Race Power Pack dimensions

Appendix vii Car deceleration system

Appendix viii Example of component identification

Appendix ix Halo and ballast container drawings

Clarifications

Help text

Please note: any amendments made prior to the event will be indicated using red underlined text.

Please read the whole document without assumptions from previous rules documents.

When you see green italic text, this is intended to help clarify a regulation or diagram.

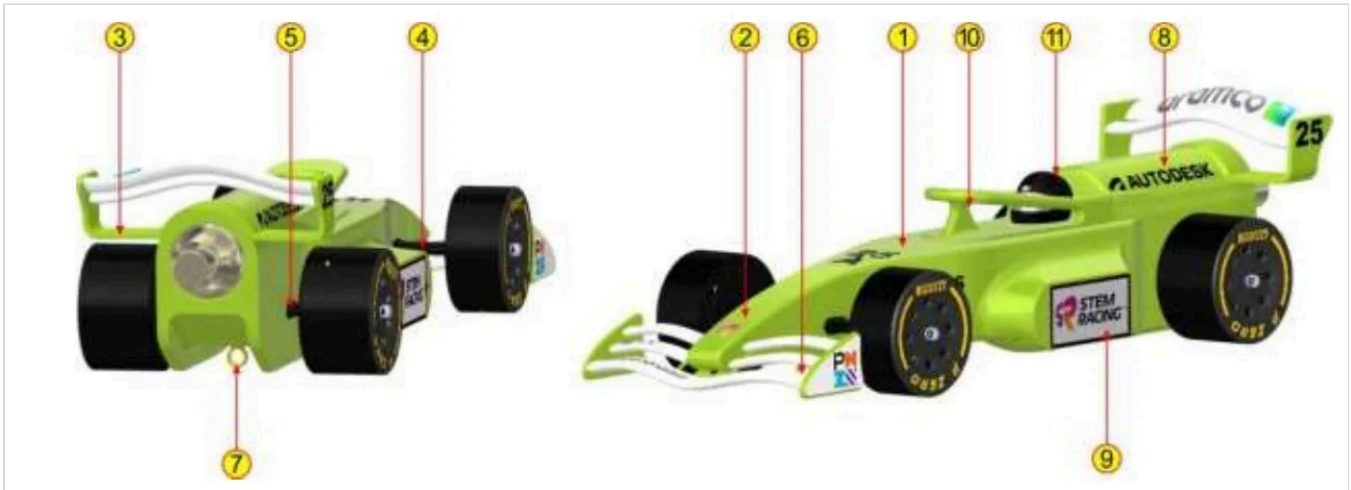
When you see the PP+ symbol in the margin, this means a rule carries a Proportional Penalty. Find out more about Proportional Penalties and the classification of rules in T2.5.

Article T1 Definitions

T1.1 STEM Racing™ India car

This is also referred to as 'the car'. Designed and manufactured according to these regulations for the purpose of participating in races on the STEM Racing™ India track at the National Finals event, powered only by a single 8 grams power unit cartridge. STEM Racing™ India cars are designed to travel the 20-metre race and withstand the physical deceleration required after crossing the finishing line.

A STEM Racing™ India car assembly must only consist of the following components:



The numbered components of the car listed in T1.1 (the legal ballast, item 12, sits inside the body under the halo).

Adhesives with no dimensional impact are permissible for joining components.

Example parts:

- A body encompassing a virtual cargo and a power unit cartridge machined from a single piece of STEM Racing™ Model Block Material

- Nose cone assembly (Optionally Replaceable)

 - Front wing support structure

 - Front wing(s)

 - End plates

Rear Wing Assembly

- Rear wing support structure

- Rear wing(s)

- End plates

Front wheel support system(s)

- Front wheel(s) rotating.

- Wheel support structure non rotating

Rear wheel support system(s)

- Rear wheel(s) rotating.

- Wheel support structure non rotating

Front tether line guide

Rear tether line guide

Surface finishing and decals

STEM Racing™ India Logo Decal (2)

Halo

Drivers Helmet

Legal Ballast

Refer to T1.22 and T4 for more information

T1.2 Fully assembled car

A STEM Racing™ India car, without a power unit cartridge inserted, presented ready for racing, resting on all four wheels on the track surface, free of any external force other than gravity.

T1.3 Body

The body is a single uninterrupted piece of STEM Racing™ Model Block material existing rear of reference plane A and encompassing both the virtual cargo and power unit cartridge chamber. For dimensional purposes the body also includes any attached decals and surface finishes. Any STEM Racing™ Model Block forward of reference plane A is not defined as body.

T1.4 Power unit cartridge chamber

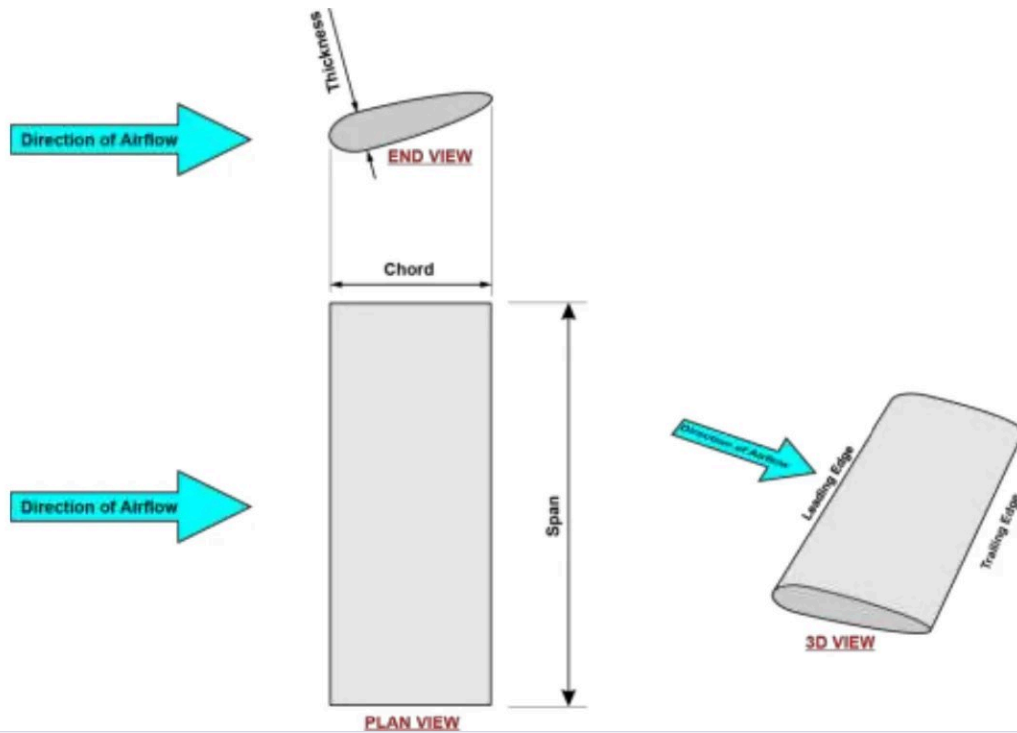
The power unit cartridge chamber is a cylindrical clear space bounded around its inner circumference and forward end by car body only. This is where the power unit cartridge is placed for racing.

T1.5 Wing

A wing on a STEM Racing™ India car is an aerodynamic feature that permits airflow around its minimum required chord surface including its features of a leading and trailing edge across its minimum required span width. A wing is dimensionally defined by the span, chord and thickness. The vertical cross-sectional shape of the wing, parallel to the direction of car travel, is referred to as an aero-foil.

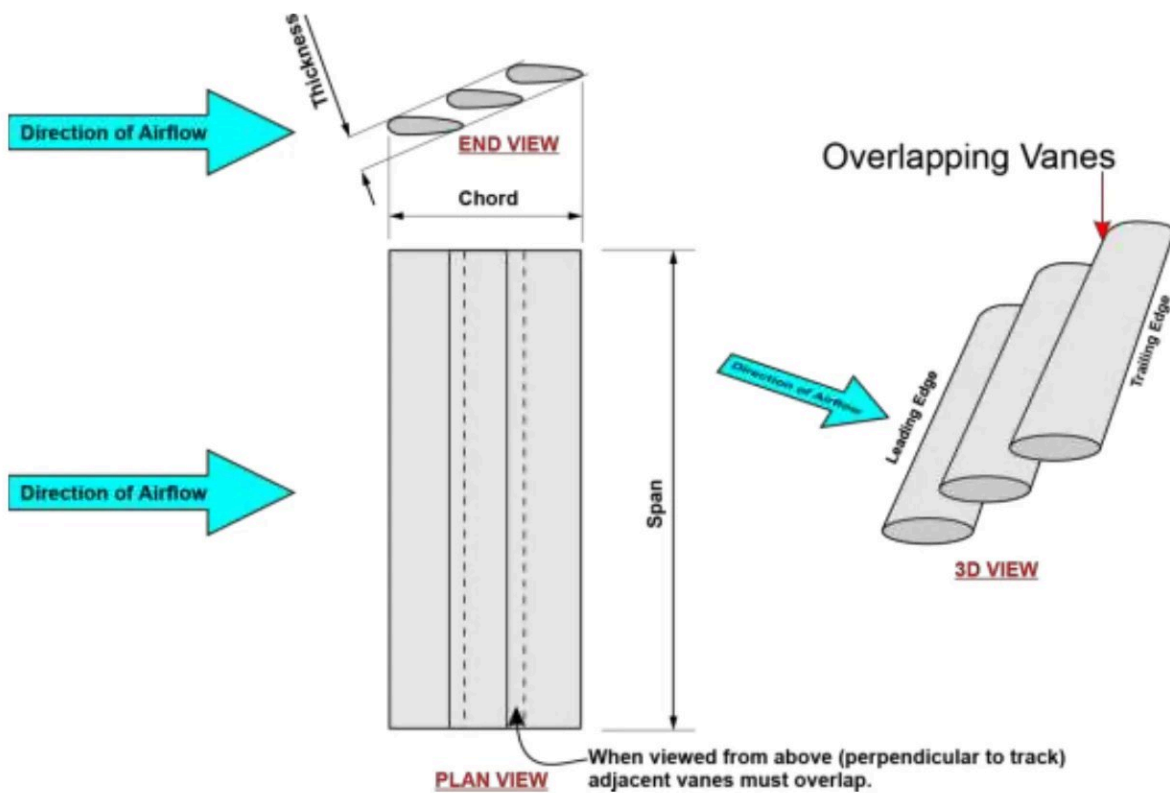
Wing cross-section / aero-foil naming terminology:

Wing naming terminology applied to single vane wing



Wing naming terminology applied to single vane wing (T1.5).

Wing naming terminology applied to multiple vane wing



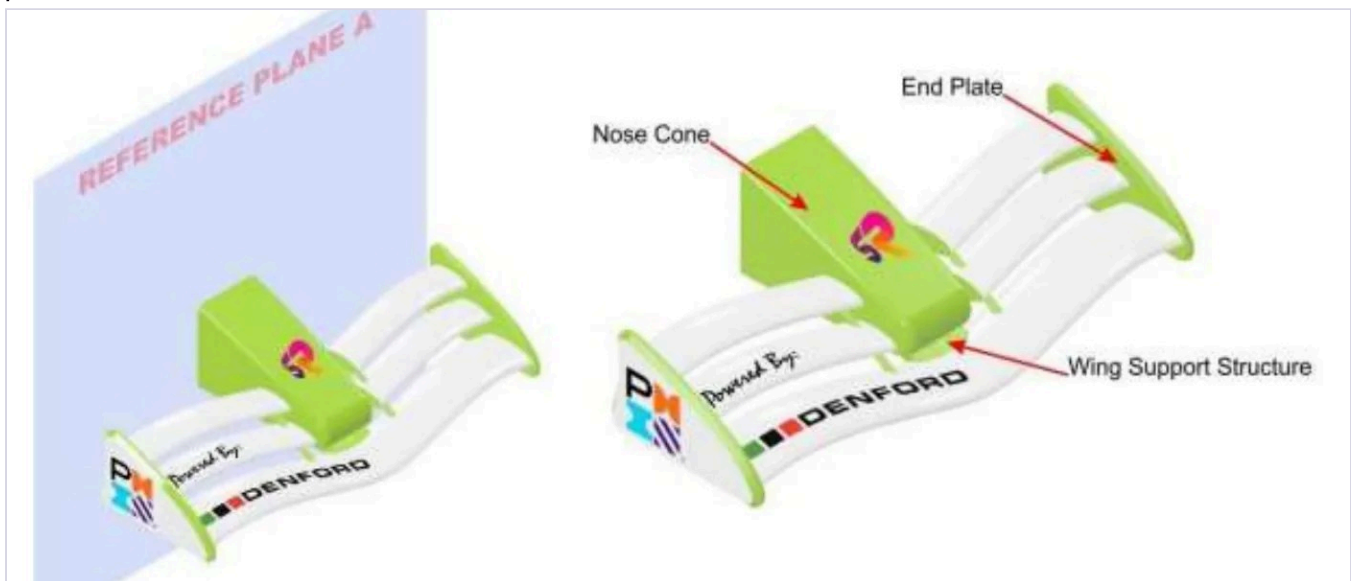
Wing naming terminology applied to multiple vane wing (T1.5).

T1.6 Wing support structure and end plate

- T1.6.1 Front Wing support Structure** Front wing support structure is a feature forward of reference plane A, that joins a wing(s) to the nose cone.
- T1.6.2 Rear Wing support Structure** Rear wing support structure is a feature rearward of reference plane B, that joins the wing to the car body.
- T1.6.3 End Plate** Any structure that forms part of a wing assembly outside of the minimum wing span.

T1.7 Nose cone

The nose cone is a component of the car, which is used as a wing support structure, that only exists forward of reference plane A. This includes any STEM Racing™ Model Block material or any other materials that continue forward of reference plane A.



The nose cone, end plate and wing support structure forward of reference plane A (T1.7).

T1.8 Wheel

A wheel is a single part or assembly of components, cylindrical in form, with its maximum circumference contacting the track surface, allowing forward motion of the car through rotation.

T1.9 Wheel support system

Wheel support systems are single parts or an assembly of components that connect a wheel to any other part of the car, they MUST be contained within a theoretical cylindrical extrusion defined by the actual diameter of the relative wheels across each axle. These may be sourced from a supplier or manufactured wholly or in part by the team.

T1.10 Tether line guide

A tether line guide is a key safety component which completely surrounds the track tether line so as to safely connect the car to the tether line during races. A tether line guide can be a component sourced from a supplier or manufactured wholly or in part by the team.

T1.11 Surface finish and decals

A surface finish on a STEM Racing™ India car is considered to be any applied visible surface covering, of uniform thickness over the profile of a STEM Racing™ India car assembly component. A decal is material adhered to a component or surface finish. To be defined as a decal, 100% of the area of the adhering side must be attached to a surface. Surface finishes and

decals are included when measuring the dimensions of any components they feature on.

T1.12 STEM Racing™ India logo decal

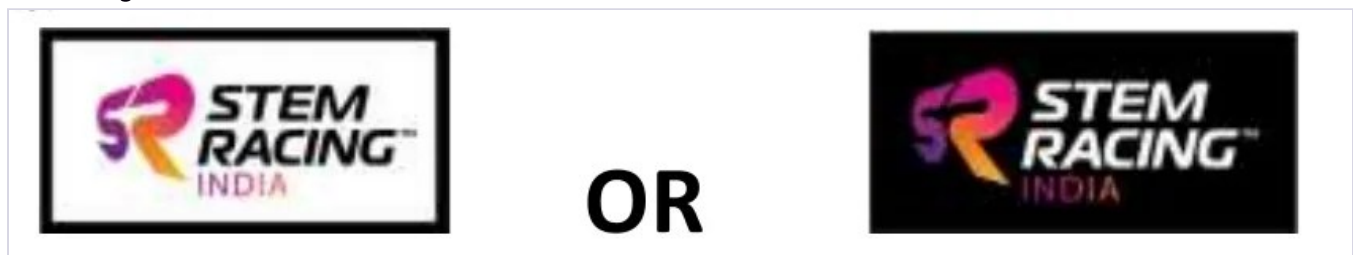
Teams must use a labelled decal to identify Car A and Car B and a blank decal to identify display car(s). The official decals are supplied by STEM Racing India at event registration.

This consists of the STEM Racing™ India logo graphic printed on a black or a white adhesive vinyl with a 1mm contrast keyline border, with a horizontal dimension of 30mm and vertical dimension of 15mm. Teams choose to use either the black or white outlined decal so as to provide maximum contrast to the surface color the decal is being adhered to.

A team can manufacture and fit their own decals, provided they use the official STEM Racing™ India logo decal artwork which can be downloaded from the below link:

https://drive.google.com/file/d/11t56pArfPVFbIFcMD7KTnzL9w_gE142e/view?usp=drive_link

Decal designs:



The two official logo decal designs (T1.12): white with a black keyline, or black.

T1.13 Hand finishing

Hand finishing is defined as use of a hand powered device (e.g. abrasive paper) for removing only the irregularities that may remain on a CNC machined surface of the car body. These irregularities are often referred to as 'machined marks', any hand finishing must be kept to a minimum.

T1.14 Official STEM RACING™ Model Block

The official STEM Racing™ Model Block Material is a rigid, closed cell foam block processed to the dimensional features as shown by diagrams in the appendix of this document.

IMPORTANT: all cars entered into the STEM Racing™ India National Finals 2025/26 must be manufactured from STEM RACING™ Model Blocks.

T1.15 Engineering drawings

Engineering drawings are CAD produced drawings which should be detailed such that, along with relevant CAM programs, could theoretically be used to manufacture the fully assembled car by a third party. Such drawings **MUST** include all relevant dimensions and material information.

Where stated, STEM Racing™ India engineering drawings of a readable scale **MUST** be clearly dimensioned and identified by **hatching, shading, block color or boundary line** within the engineering drawings to specifically identify and prove compliance for the virtual cargo, front and rear wheel support structures, nose, front and rear wing support structures and front and rear wing surfaces.

Clearly labelled **construction or boundary lines** are acceptable to define the boundaries between components such as nose cone, wheel support and wing support structures.

Engineering drawings can include: orthographic projection, auxiliary projection, section views, isometric projection, oblique projection, perspective and annotated renderings.

It is recommended to label the relevant technical regulations where appropriate (e.g T9.5.1: 25.0mm) throughout your Engineering drawings; this makes the job of the scrutineer much easier in identifying the different features of your car.

See competition regulations scorecard for mandatory detailed list.

T1.16 Renderings

Renderings are images intended to illustrate the three-dimensional form of an object. These can be generated in isometric projection, oblique projection, or perspective.

T1.17 Reference planes

T1.17.1 Vertical reference plane

To assist with describing dimensions, it is assumed that a two-dimensional invisible plane exists along the length of the power unit cartridge chamber center axis and normal to the track surface. This is known as the vertical reference plane.



The vertical reference plane (T1.17.1).

T1.17.2 Reference plane A

To assist with describing dimensions, it is assumed that a two-dimensional invisible plane exists 16.0mm in front of the front axle center line normal to the track surface. This is known as reference plane A.

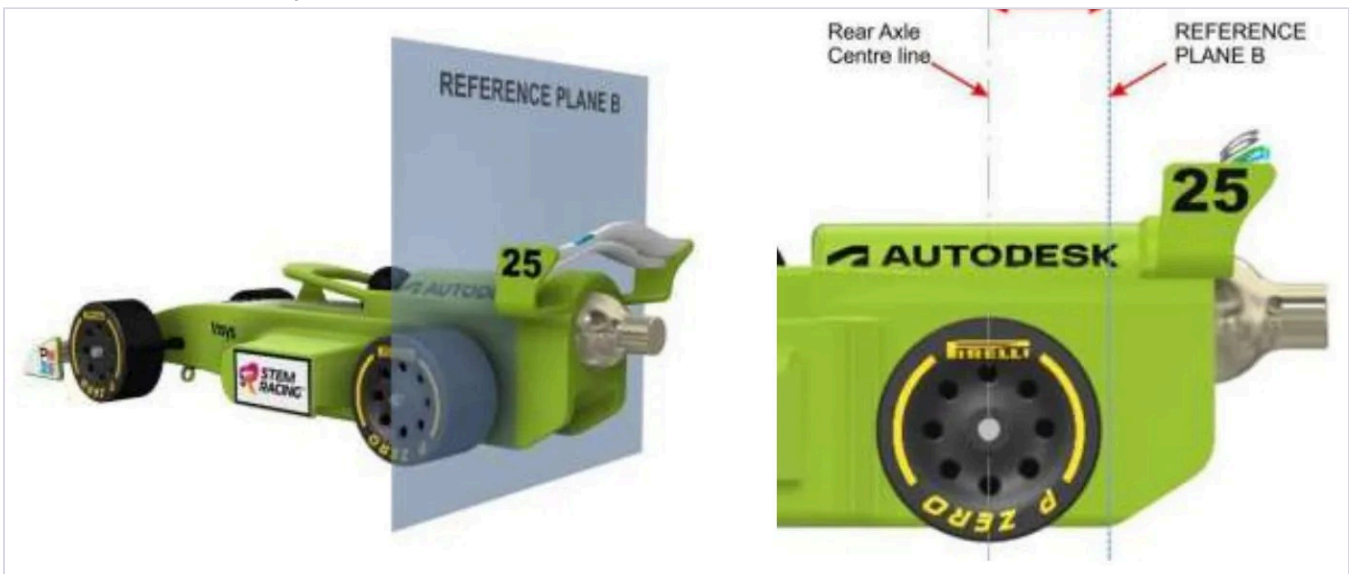


Reference plane A, 16.0mm in front of the front axle center line (T1.17.2).

T1.17.3 Reference plane B

To assist with describing dimensions, it is assumed that a two-dimensional invisible plane exists 16.0mm to the rear of the rear axle center line normal to the track surface.

This is known as reference plane B.



Reference plane B, 16.0mm to the rear of the rear axle center line (T1.17.3).

T1.18 Additional components

Any component other than those listed in T1.1 will be considered an additional component.

T1.19 Normal

The term 'normal' can be used in geometry to describe a line or object that is perpendicular or at 90 degrees to another given object. When referring to the term normal in these regulations it is considered to mean:

Being at right angles; perpendicular.

Perpendicular to the direction of a tangent line of a curve or a tangent plane to a surface.

T1.20 Full 8-gram power unit cartridge

A metallic cartridge which contains an 8-gram charge of compressed air. For weights and dimensions refer to appendix v. Official Denford Race Power Pack dimensions.



The 8-gram power unit cartridges as supplied (T1.20).

T1.21 Halo

The halo is a driver crash-protection system used in open-wheel racing series, which consists of a curved bar placed to protect the driver's head. The Halo is being introduced to STEM Racing™ India not only to echo real Formula 1® car design but also as a component of a new car deceleration system.

The Halo is available to download as a universal 3D part from the below link:

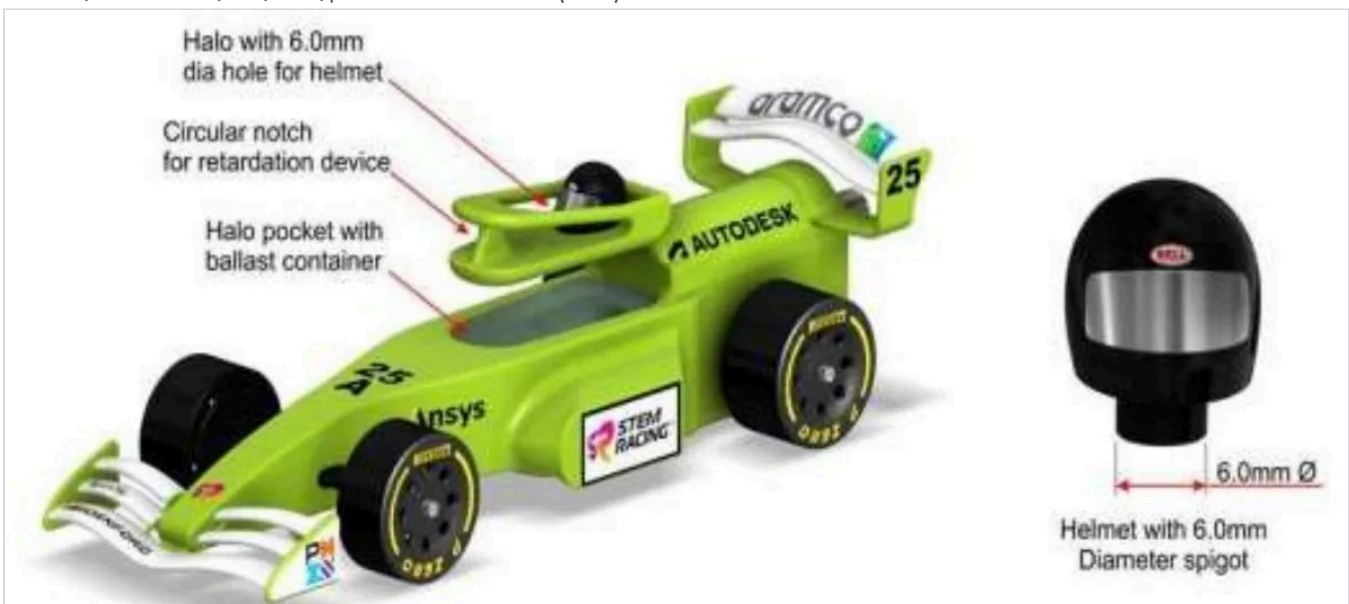
https://drive.google.com/file/d/1LqT9k6uSwtxeE2QIlyHkdd78VseXGfTt/view?usp=drive_link

The Halo MUST be included in the final car design. Detailed Engineering Drawings are available in appendix ix.

HALO – IMPACT PROTECTION SYSTEM



The Halo, shown in rear, side, front, plan and isometric views (T1.21).



Halo, halo pocket with ballast container and helmet with 6.0mm diameter spigot (T1.21).

The Halo must be well adhered to the body of the car, to achieve this requires a recess or pocket to match the underside of the Halo in the car body. It has been designed so the required recess or pocket can be manufactured using a 1/4" (6.35mm) ball nose cutter. The Halo and the Helmet are mandatory items and can be made from any material. Please refer to T4.4 for the halo file. Please see Appendix ix for further details.

T1.22 Legal ballast

To help achieve the minimum weight, legal ballast may now be added only to the container under the halo, accessible through the halo/helmet aperture (hole). This is the only permitted location for ballast. The ballast can consist of materials such as fishing lead weights or similar

Article T2 General principles

T2.1 Regulations documents

T2.1.1 STEM Racing India, issues the regulations, their revisions and amendments made.

T2.1.2 Technical Regulations - this document. The Technical Regulations document is mainly concerned with those regulations that are directly related to STEM Racing™ India car design and manufacture. Technical regulation article numbers have a 'T' prefix.

T2.1.3 Competition Regulations – a document separates to this one which is mainly concerned with regulations and procedures directly related to judging and the competition event. Competition Regulation article numbers have a 'C' prefix.

T2.2 Interpretation of the regulations

T2.2.1 The text of a regulation and any related definitions should be considered together for the purpose of interpretation.

IMPORTANT: Diagrams and or images are for illustration purposes only and do not contribute to regulatory compliance.

T2.2.2 Text clarification - any questions received that are deemed by STEM Racing India to be related to regulation text needing clarification will be answered by STEM Racing India. The question received, along with the clarification provided by STEM Racing India, will be published to all competing teams at the same time.

T2.3 Amendments to the regulations

Any amendments will be announced and released by STEM Racing India by email notification to all the Teams. Any amended text will be indicated thus (using red underlined text).

T2.4 Classification of regulations

T2.4.1 The technical regulations are classified as either: **GENERAL, SAFETY, PERFORMANCE.**

GENERAL	SAFETY	PERFORMANCE
Regulations that shape the way the car fundamentally looks and works, vital to the style of a STEM Racing™ car.	Mandatory rules that govern the safe running of the car. Cars must meet these rules to be considered 'safe to race'.	Rules that have a direct impact on the performance of the vehicle, these typically carry the heaviest penalties.

T2.4.2 If a race car is judged as being NON-COMPLIANT with any Performance regulation they will be INELIGIBLE for the awards of: 'Fastest Car' and 'Best Engineered Car'. All Performance regulations are highlighted in yellow throughout this document.

T2.5 Compliance with regulations

T2.5.1 Points are deducted for non-compliance with the technical regulations as per the penalties as defined in this document. Both race cars are assessed during Specification Judging and points will be deducted for any infringements on either car. These penalties are only applied once, per infringement, per car. After initial Specification Judging any team with “safety” infringements will be given a 10-minute service session. See competition regulations for more information.

T2.5.2 Proportional penalties will be applied to the following regulations:
T3.4, T3.5, T3.6, T3.7, T5.2, T5.6, T7.2, T7.3, T7.4, T7.5, T7.11, T8.6, T8.6.2, T8.6.3, T9.5.1, T9.5.2, T9.5.3, T9.5.4
 The penalty applied increases proportionally as the margin of non-compliance with the absolute minimum/maximum dimension increases by rounding up the non-compliance to the next complete unit of measure (1.0mm or 1.0g). The penalty is applied once for every complete unit outside of the absolute minimum/maximum dimension. For example:



Worked example of a proportional penalty: T9.5.1 Rear wing span, PERFORMANCE, penalty 2pts per millimeter (T2.5.2).

T2.6 Design ideas and regulation compliance questions

Teams are not permitted to seek a ruling from STEM Racing India or any competition officials or judges before the event as to whether a design idea complies with these regulations, any regulatory communication between teams and ICCs or teachers is only for guidance, rulings will only be made by the official judges at the National Finals event. Design compliance to the regulation's forms part of the competition. As in Formula 1® innovation is encouraged and STEM Racing™ India teams may also find ways of creating design features that push the boundaries of the regulations in order to get an extra competitive edge.

T2.7 Measurements

T2.7.1 All dimensions and weights are presented as absolute minimum or maximum, unless stated otherwise. For example:

MIN WEIGHT	MIN DIMENSION	MAX DIMENSION
Absolute Min: 50.0g	Absolute Min: 26.0mm	Absolute Max: 34.0mm
50.0g - PASS	26.0mm - PASS	34.0mm - PASS
49.9g - FAIL	25.9mm - FAIL	34.1mm - FAIL

***Note – Important regulations for Main Car Body (excluding nose cone & rear wing) –**

The **Main Car Body**, including the support structure, must have a maximum overall length of 205mm to meet the required design specifications and competition standards.

Absolute Max: 205.0mm

The **Main Car Body** must maintain a minimum thickness of 3.5mm at every point to ensure structural integrity and compliance with competition regulations.

Absolute Min: 3.5mm

T2.7.2 **Dimensional measures** - all car component dimensions or weight are inclusive of any applied paint finish or decal. A series of specially manufactured gauges will be used to broadly verify dimensional compliance. Accurate measuring tools, such as vernier calipers, will then be used to closely inspect any dimensions found to be close to the dimensional limits per the initial gauge inspection. **IMPORTANT:** Some regulations are assessed with a full 8g race cartridge fully inserted into the cartridge chamber. For compliance with these regulations, the **static weight distribution** of the car must be such that the car is capable of **resting on all four (4) wheels** without any outside assistance.

T2.7.3 **Weight measures** – all weight measurements will be made using the STEM Racing India calibrated electronic competition scales.

T2.8 Benefit of doubt

The chair of judges will, where appropriate, seek to use 'benefit of doubt' when the assessment of compliance is marginal or unclear. In this situation, teams will be given the benefit of doubt rather than a firm penalty if a penalty cannot be clearly measured or identified.

T2.9 Spirit of the competition

Teams are expected to act in the spirit of the competition, both before and during the STEM Racing™ India National Finals. Any team or individual team member deemed by the chair of judges to be acting outside of the spirit of the competition, can be removed from certain or all aspects of the competition. For example, a team or individual team member attempting to abuse the technical regulations to their advantage may, at the discretion of the chair of judges, be removed from racing and receive no points for this activity. A team or individual team member deemed to be acting in an unsportsmanlike manner towards another team or other persons may be removed from some or all judging areas.

The spirit of the competition is simple; embrace and respect the rules and regulations, do your very best to compete legally and fairly, while contributing positively to the STEM Racing™ India National Finals. Make friends, create positive relationships, network professionally and enjoy yourselves.

T2.10 Originality of work

STEM Racing India welcomes and endorses innovation and does not consider that plagiarism should play any part in any of the disciplines that make up the competition.

Competing teams at all levels of the competition that intentionally plagiarize any part of their assessed work, undermines the credibility and integrity of the STEM Racing™ India challenge and the spirit of the competition.

At the STEM Racing™ India National Finals 2025/26, STEM Racing™ India shall be implementing various originality detection methodologies and requesting all competing teams to submit an originality declaration. Further details and associated penalties will be explained in detail in the STEM Racing™ India National Finals 2025/26 competition regulations.

Article T3 Fully assembled car

T3.1 Design and manufacture

[GENERAL | Penalty – 5pts each]

- T3.1.1** All STEM Racing™ India cars must be designed and engineered using CAD (Computer Aided Design) and CAM (Computer Aided Manufacture) technology. CAD software used should provide for 3D part modelling, assembly and 3D realistic rendering. We recommend the use of Autodesk Fusion 360 software. The CAM package should allow students to simulate CNC machining processes so they can show evidence of these in their portfolio. We recommend the use of Denford QuickCAM PRO software.
- T3.1.2** The body of all STEM Racing™ India cars must be manufactured via material removal using a CNC router/milling machine. We recommend all teams use a Denford CNC Router. This manufacturing process should occur at your school/college or at a designated manufacturing center/partner site.
- T3.1.3** An official STEM Racing™ holographic sticker from the official STEM Racing™ Model Block for each car must be submitted on the project element submission sheet at the National Finals event registration. STEM Racing India stickers will also be permitted.
- T3.1.4** The individual components of both race cars must be designed with identical geometry.

T3.2 Safe construction

[SAFETY | Penalty – 10pts each]

- T3.2.1** **Specification judging** - all submitted cars will be inspected closely to ensure that they are engineered and constructed safely for the purpose of racing. If the judges rule an aspect of either race car to be unsafe for racing, the team will be required to carry out repairs / modifications to the car(s). Any such repair work will result in a penalty of 10 points per unsafe car.
- T3.2.2** **During racing** – the race officials will routinely inspect cars for safety during scheduled races. If the officials rule a car to be unsafe, a penalty of 10 points will be imposed at the discretion of the Chair of Judges. The team may repair the car as per the Competition Regulations – C10 Car Repairs and Servicing.

T3.3 Defined features

[PERFORMANCE | Penalty – 20pts]

The car assembly must only consist of components listed in ARTICLE T1.1.

T3.4 Total width

[GENERAL | Penalty – 5pts per millimeter]

Total width is the maximum assembled car width, measured normal to the vertical reference plane, between the outer edges of the widest feature of the car assembly.

Absolute Min: 65.0mm / Absolute Max: 85.0mm

T3.5 Total height

[GENERAL | Penalty – 5pts per millimeter]

Total height is the maximum assembled car height, normal to the vertical reference plane, between the track surface and the highest feature of the car assembly. This is measured with a full 8g power unit cartridge inserted into the cartridge chamber with the car sitting on all four (4) wheels with no outside assistance.

Absolute Max. 65.0mm



Total height (T3.5, max 65.0mm) and total width (T3.4, max 85.0mm).

T3.6 Total weight

[PERFORMANCE | Penalty – 10pts per gram]

Total weight is the weight of the car excluding a power unit cartridge. If ruled underweight at car registration and confirmed during initial specification judging using the official competition scales, the above points penalty will be applied. No car will race underweight and ballast will be added to the underside of the car at 0.2g for every 0.1g underweight.

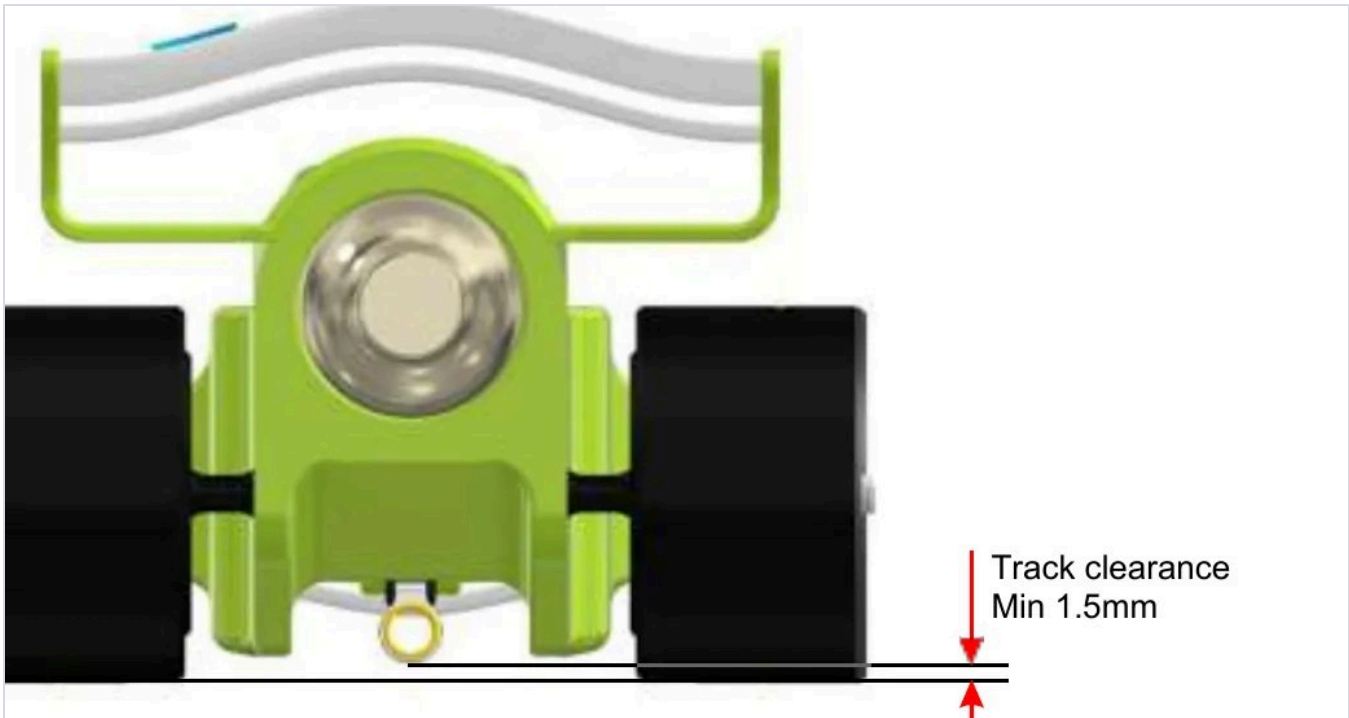
Absolute Min: 50.0g

T3.7 Track clearance

[GENERAL | Penalty – 10pts per millimeter]

Track clearance is the distance between track surface and any car component as listed in T1.1, except wheels. Measured normal to the track surface. This is measured with a full 8g power unit cartridge inserted into the cartridge chamber with the car sitting on all four (4) wheels with no outside assistance.

Absolute Min: 1.5mm



Track clearance, minimum 1.5mm (T3.7).

T3.8 Status during racing

[GENERAL | Penalty – 5pts]

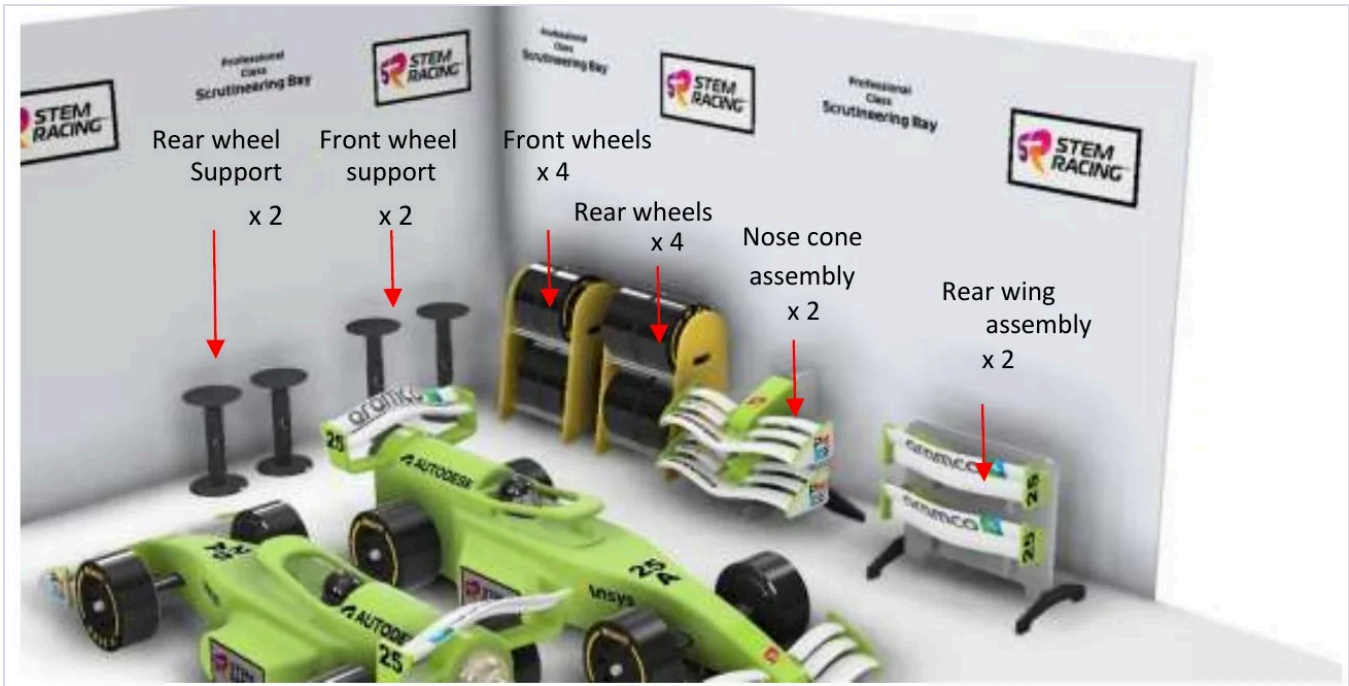
The car assembly must be designed so that no items other than those listed in T3.9, or power unit cartridges are removed, replaced or added to the assembly during scheduled race events.

T3.9 Replacement components

[GENERAL]

Any spare / replacement components should be identical in design and geometry to those fitted to the car and must be submitted with the cars at registration. Only the following spare / replacement components are permitted:

COMPONENT	MAX QUANTITY
Nose cone & front wing assembly	2
Rear wing assembly	2
Front wheels	4
Front wheel support structure	2
Rear wheels	4
Rear wheel support structure	2



The permitted spare and replacement components (T3.9).

- T3.9.1 Submitted components – [GENERAL]** Only submitted replacement components that are determined by the judges to be identical in design and geometry to those fitted to the cars will be allowed to be used during track repairs or post-race servicing.
- T3.9.2 Introduced components - [GENERAL | Penalty – 5pts per component]** If any other items need to be introduced for car repairs a 5pt penalty will be incurred per component item.

Article T4 Body

T4.1 Body construction

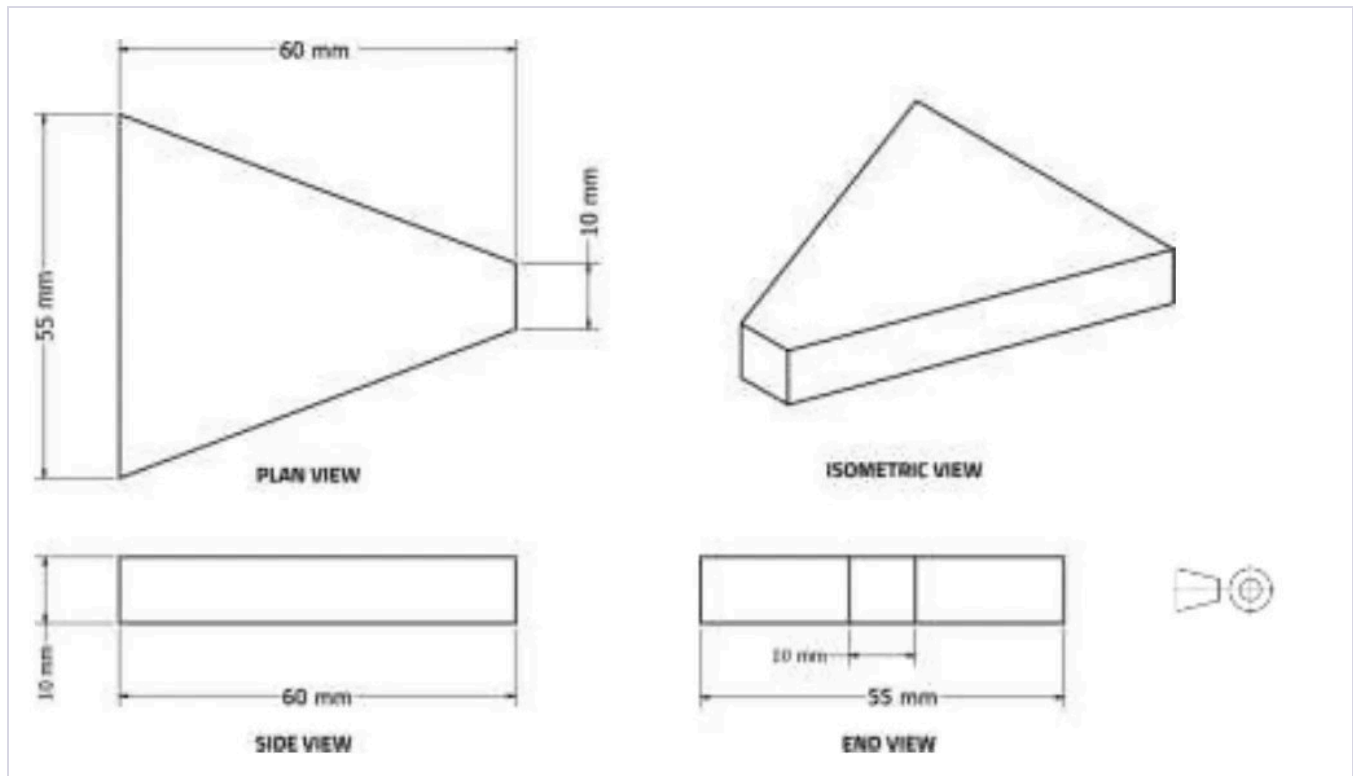
[GENERAL | Penalty – 20pts]

A single continuous piece of CNC manufactured STEM Racing™ Model Block material must exist rear of the reference plane A, encompassing both the virtual cargo and power unit cartridge chamber.

T4.2 Virtual cargo

[PERFORMANCE | Penalty – 25pts]

A virtual cargo must be completely encompassed by the body and be wholly positioned between the front and rear wheel center lines. The virtual cargo must have minimum dimensions as shown below, with its top surface located symmetrically about and positioned normal (90 degrees) to the vertical reference plane. The virtual cargo may also share common faces with the car body. All dimensions shown are absolute minimum. The Virtual Cargo can coincide with the legal ballast container but not the halo pocket.

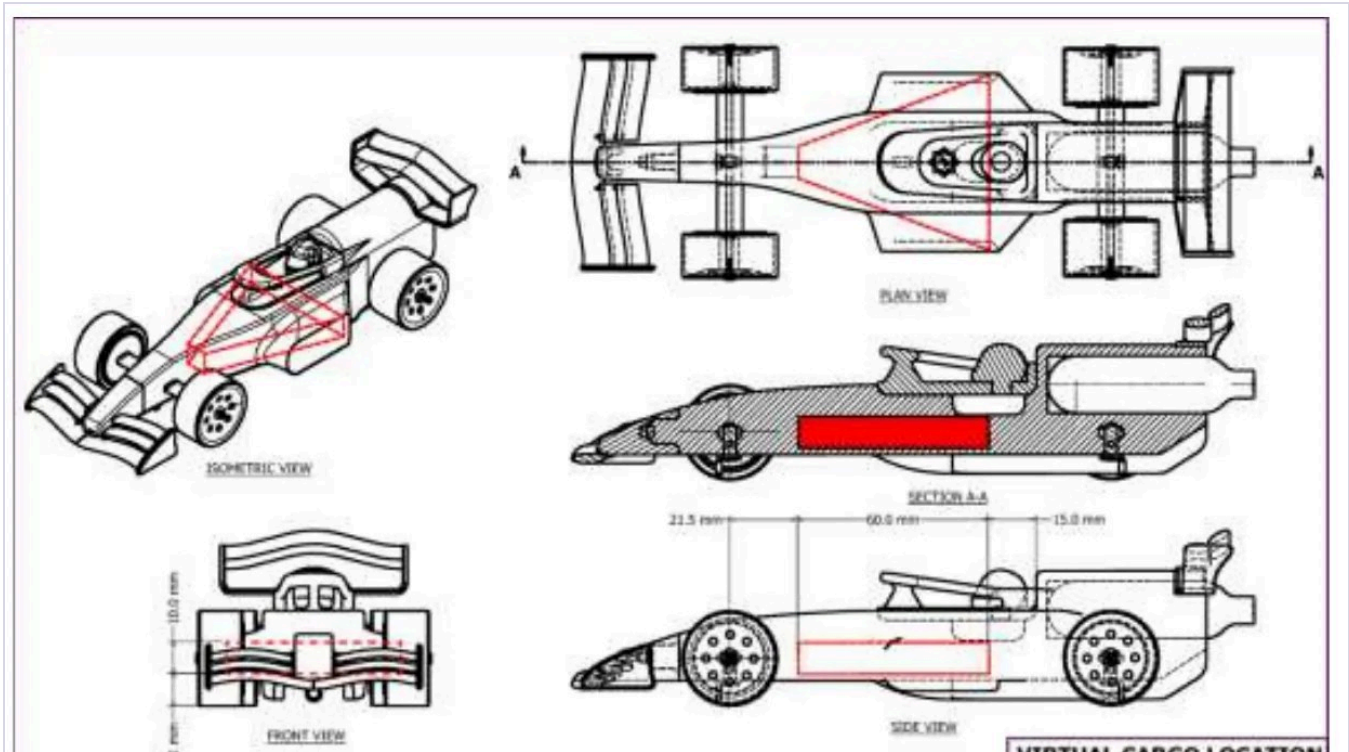


Minimum dimensions of the virtual cargo (T4.2): 60 mm x 55 mm tapering to 10 mm, 10 mm high.

T4.3 Virtual cargo identification

[GENERAL | Penalty – 5 pts]

The virtual cargo location and compliance **MUST** be clearly dimensioned (such as from either the front or rear axle center lines and height above the track). The virtual cargo **MUST** be identified by hatching, shading or block color, or outlined with a contrasting color line within the engineering drawings submitted for scrutineering. Please refer to the example diagram below, showing the virtual cargo clearly highlighted in red:



Example engineering drawing with the virtual cargo highlighted in red (T4.3).

T4.4 Halo

T4.4.1

Halo – [PERFORMANCE | Penalty – 10 pts] The Halo (refer ARTICLE T1.21) MUST be included in the car design without any dimensional or surface profile changes. The file can be downloaded from https://drive.google.com/file/d/1LqT9k6uSwtxeE2QIlyHkdd78VseXGfTt/view?usp=drive_link Please see appendix ix for detailed dimensions.



The Halo part and its pocket in the body (T4.4.1).

T4.4.2

Halo visibility front and side views – [GENERAL | Penalty – 10 pts] Visibility of the Halo must not be physically obstructed by any other component when viewed in the front or side views.

SIDE VIEW When viewed from the side, everything inside the red outline MUST be visible.

FRONT VIEW When viewed from the front, everything inside the red outline MUST be visible.



Halo visibility in front and side views (T4.4.2). The printed labels above the two views read SIDE VIEW then FRONT VIEW.

T4.4.3

Halo visibility top view – [GENERAL | Penalty – 10 pts] The Halo must not be physically obstructed in the plan view except by the helmet.

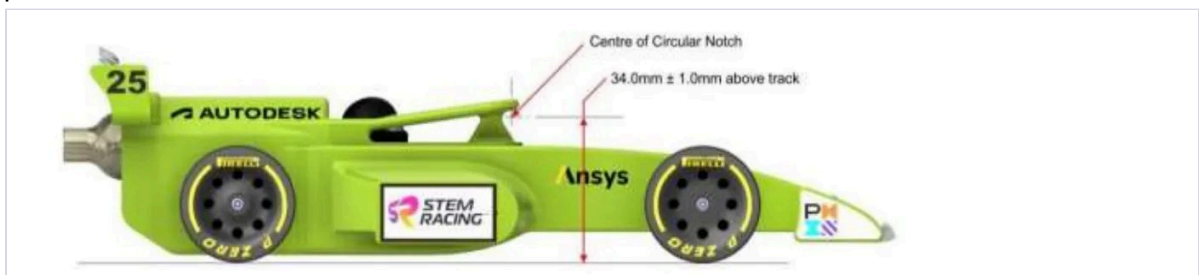
TOP VIEW When viewed from the top, everything inside the red outline MUST be visible.



Halo visibility in the top view (T4.4.3).

T4.4.4

Halo circular notch height– [SAFETY | Penalty – 5pts] To be effective the center of the Circular Notch must be 34.0mm (± 1.0 mm) above the track surface. To achieve this the bottom of the halo pocket must be 24.0mm and parallel to the track surface in the assembled car.



Height of the centre of the circular notch above the track (T4.4.4).

T4.4.5

Halo Safety Test– [SAFETY | Penalty – 5pts] With the car supported on a power unit cartridge a 1kg load will be suspended to give a loading of 2kg (to be reviewed) at point of contact on the Halo circular notch.



The halo safety test rig (T4.4.5).

T4.5 Helmet

[GENERAL | Penalty – 5pts]

The Helmet is a standard part designed by STEM Racing™ that MUST be included in the car design without any dimensional changes. The Helmet is available to download as a universal 3D part from the below link:

https://drive.google.com/file/d/1U3kd9Qh7LR_hv_htGgKA1zm3EyGKUKOs/view?usp=drive_link

The helmet may be manufactured out of any material.

We suggest that the helmet be painted in the team's country or hero driver colors



The helmet and its spigot: 6.0 mm diameter, 4.0 mm long (T4.5).

T4.6 STEM Racing™ India logo decal location

[GENERAL | Penalty – 5pts]

A STEM Racing™ India logo decal (refer ARTICLE T1.12) must be wholly adhered to each side of the car, and be 100% visible in the respective side view. Teams may manufacture their own decals but must use the artwork supplied by STEM Racing India.



Location and size of the logo decal (T4.6).

T4.7 Team number

[GENERAL | Penalty – 2pts]

After registering for the National Finals event each team shall be issued with their official team number. This must be displayed on the car body between the front edge of the halo and reference plane A. The number should be clearly visible in the plan view and a minimum text height of **8mm**. If a team submits their car without the team number, then one will be provided at the world finals event.

Teams may additionally include the team number elsewhere on the car such as the rear wing endplates.

Minimum text height: 8.0mm



Minimum text height of the team number, 8.0mm (T4.7).

T4.8 Decal thickness

[GENERAL | Penalty – 5pts]

This is measured as the total thickness of any decal, excluding any surface finishes.

Absolute Max: 0.5mm

Article T5 Power unit cartridge chamber

T5.1 Diameter

[SAFETY | Penalty – 5pts]

This is the diameter of the power unit cartridge chamber, measured at any point through its minimum depth.

Absolute Min: 18.0mm / Absolute Max: 18.5mm

T5.2 Distance from track surface

[GENERAL | Penalty – 5pts per millimetre]

This is measured with a full 8g power unit cartridge inserted into the cartridge chamber, from the rear center of the power unit cartridge to the track surface, measured normal to the track surface with the car sitting on all four (4) wheels with no outside assistance.

Absolute Min: 30.0mm / Absolute Max: 40.0mm



Distance from the track surface to the rear center of the cartridge (T5.2).

T5.3 Depth

[SAFETY | Penalty – 5pts]

The depth of the chamber is measured parallel to the vertical reference plane anywhere around the chamber circumference from the opening to the chamber end.

Absolute Min: 45.0mm / Absolute Max: 58.0mm

T5.4 Max angle of chamber

[SAFETY | Penalty – 5pts]

The absolute maximum angle of the chamber, parallel to the track surface. This is measured with a full 8g power unit cartridge inserted into the cartridge chamber with the car sitting on all four (4) wheels with no outside assistance.

Absolute Min: -3° Absolute Max: 3°

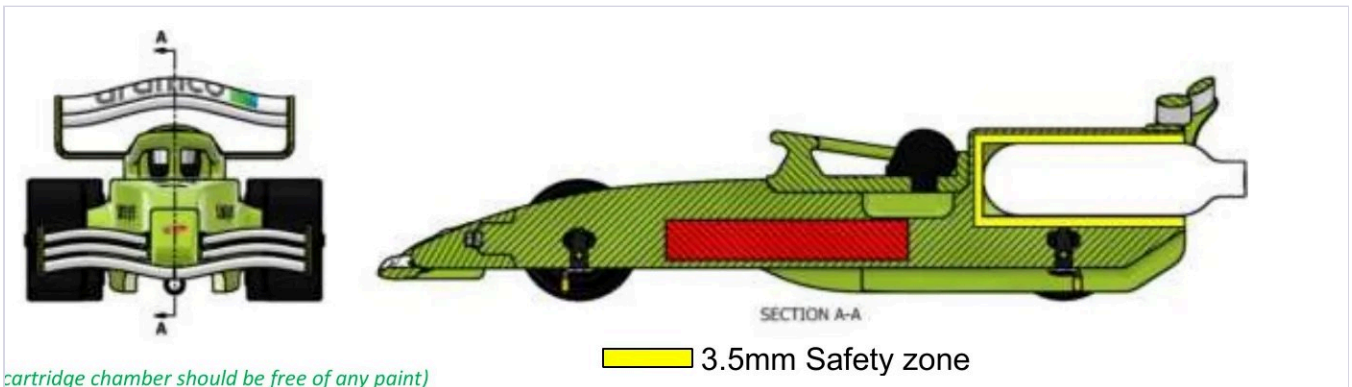
T5.5 Chamber safety zone

[SAFETY | Penalty – 10pts]

A safety zone of STEM Racing™ Model Block material with a minimum thickness of 3.5mm must be maintained around the minimum chamber depth (see T5.3). The chamber safety zone and connection to the car body will be assessed and if determined below the minimum thickness, may be considered a safety issue at the judge's discretion, see ARTICLE T3.2.

IMPORTANT: the entire circumference and depth of the power unit cartridge chamber must not be intersected by any object.

Absolute Min: 3.5mm



The 3.5mm safety zone around the chamber (T5.5).

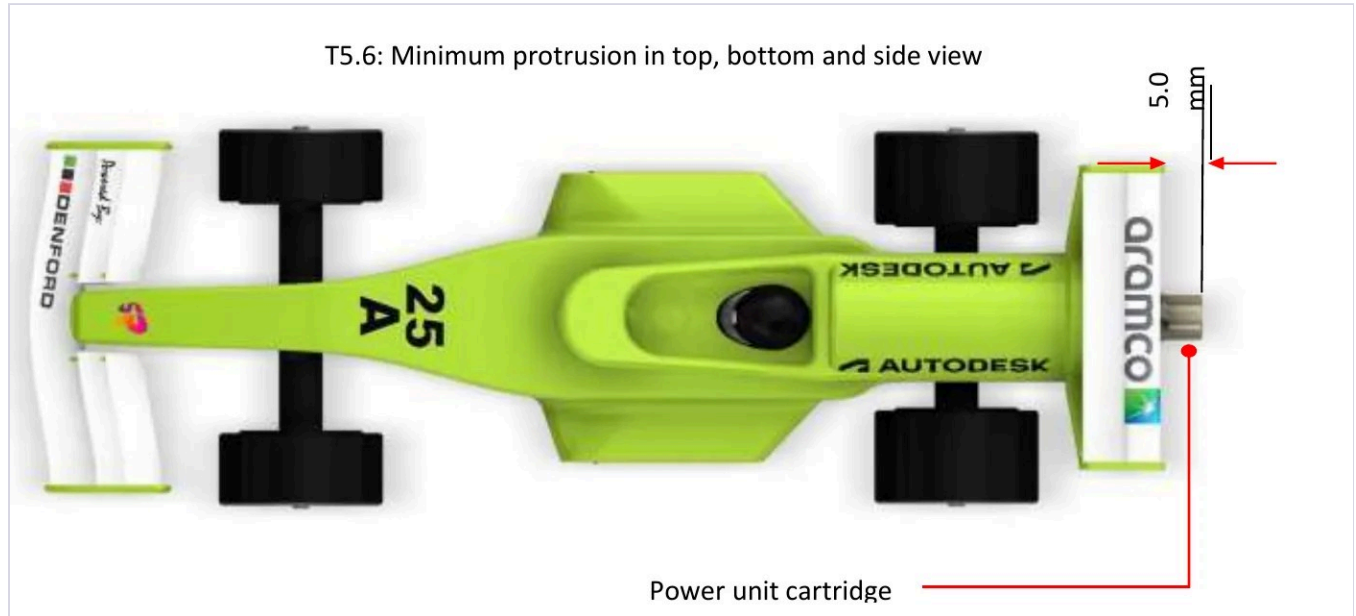
(The cartridge chamber should be free of any paint)

T5.6 Power unit cartridge visibility

[PERFORMANCE | Penalty – 10pts per millimeter]

When fully inserted, the power unit cartridge must protrude a minimum of 5.0mm from the rear of the car and be visible in its entirety when viewed radially around the longitudinal centerline of the power unit cartridge.

Absolute Min: 5.0mm



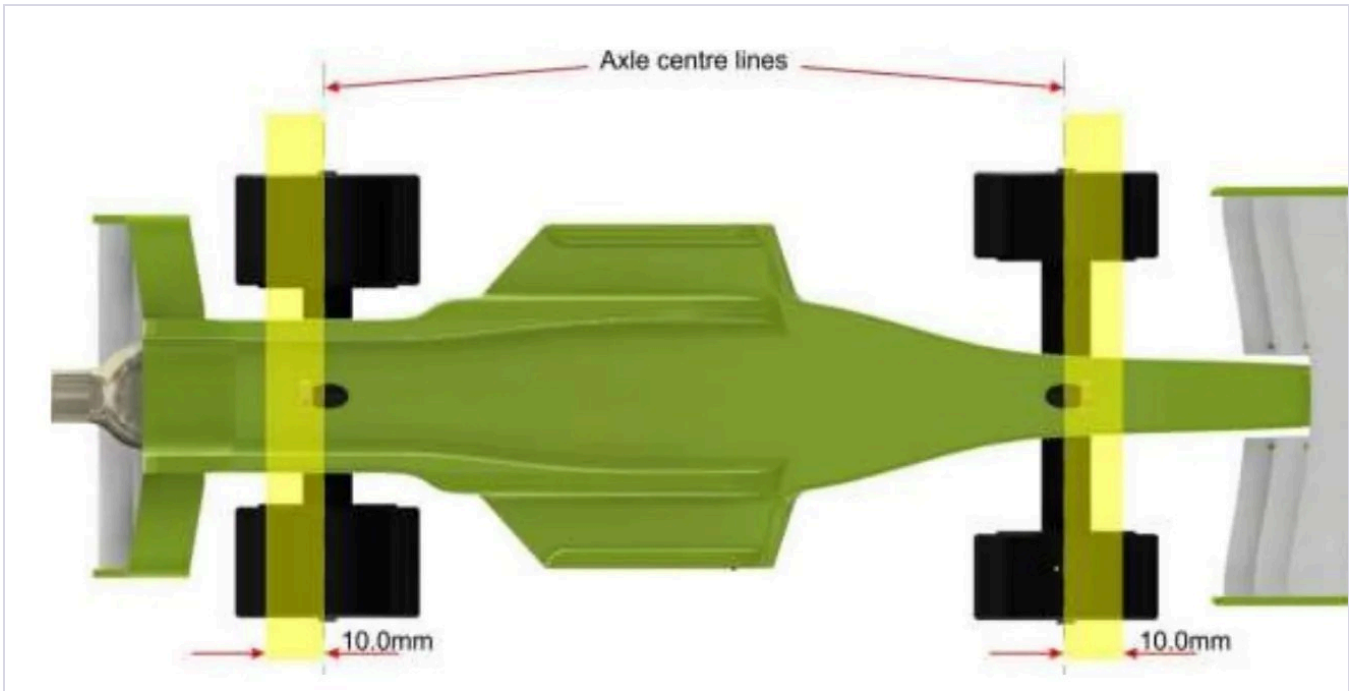
Minimum protrusion of the power unit cartridge, 5.0mm (T5.6).

Article T6 Tether line guides

T6.1 Location

[SAFETY | Penalty – 10pts]

Each car must have only two (2) firmly secured tether line guides, one on or up to 10.0mm in front of the front axle center line and one on or up to 10.0mm behind the rear axle center line of the car. The track tether line must **only** pass through the two tether line guides during racing.



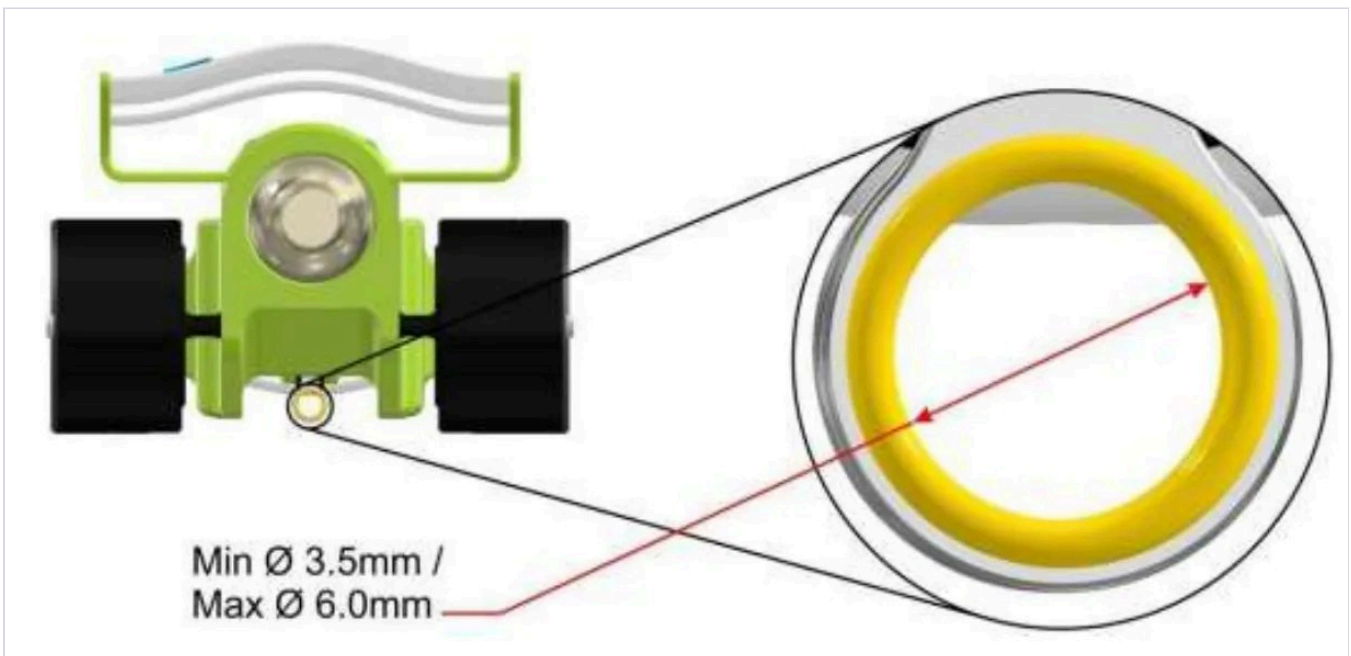
The 10.0mm zones for the two tether line guides (T6.1).

T6.2 Internal dimension

[SAFETY | Penalty – 5pts]

This is the minimum and maximum internal measurement of the opening within the guide, which the tether line passes through. This will be measured using a 3.5mm and/or 6.0mm diameter tool.

Absolute Min: 3.5mm / Absolute Max: 6.0mm



Internal diameter of the tether line guide, 3.5mm to 6.0mm (T6.2).

T6.3 Tether line guide safety

[SAFETY | Penalty – 10pts]

The guide holes must be completely closed to prevent the tether line from slipping out during racing. The construction of the tether line guides will be closely examined in relation to safety, please refer to ARTICLE T3.2 for more information. The guides must be robust so as to prevent the diameter or shape changing during racing. The below tether line guide test will be conducted during scrutineering. A 200g weight will be suspended from each tether line guide to check the guides are securely fitted to the car and safe to race.



The tether line guide test with a 200g weight on each guide (T6.3).

Article T7 *Wheels and wheel support structures*

T7.1 Number and location

[GENERAL | Penalty – 25pts]

The car assembly must include four (4) cylindrical wheels, a maximum of two (2) at the front and a maximum of two (2) at the rear. Opposing wheels must share a common center line / axis.

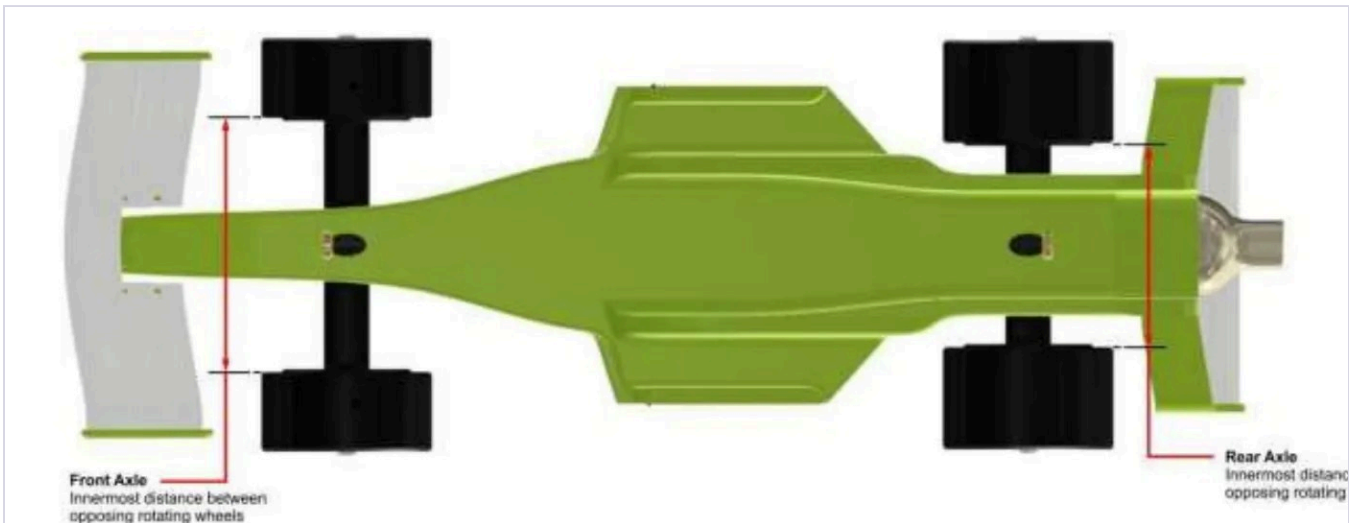
T7.2 Distance between opposing wheels

[PERFORMANCE | Penalty – 2.5pts per millimeter per axle]

This is measured as the innermost distance of the rotating wheel components (other than a rotating axle) between the two (2) opposing wheels measured parallel to the track surface.

T7.2.1 Front Axle - Absolute Min: 38.0 mm

T7.2.2 Rear Axle - Absolute Min: 30.0 mm



Distance between opposing wheels on the front and rear axle (T7.2).

T7.3 Wheelbase

[PERFORMANCE | Penalty –5pts per millimeter]

The wheelbase of the vehicle is the distance between the center line of the front and rear wheels as viewed in the side view.

Absolute Min: 120.0mm / Absolute Max: 140.0mm



Wheelbase, 120.0mm to 140.0mm (T7.3).

T7.4 Track contact width

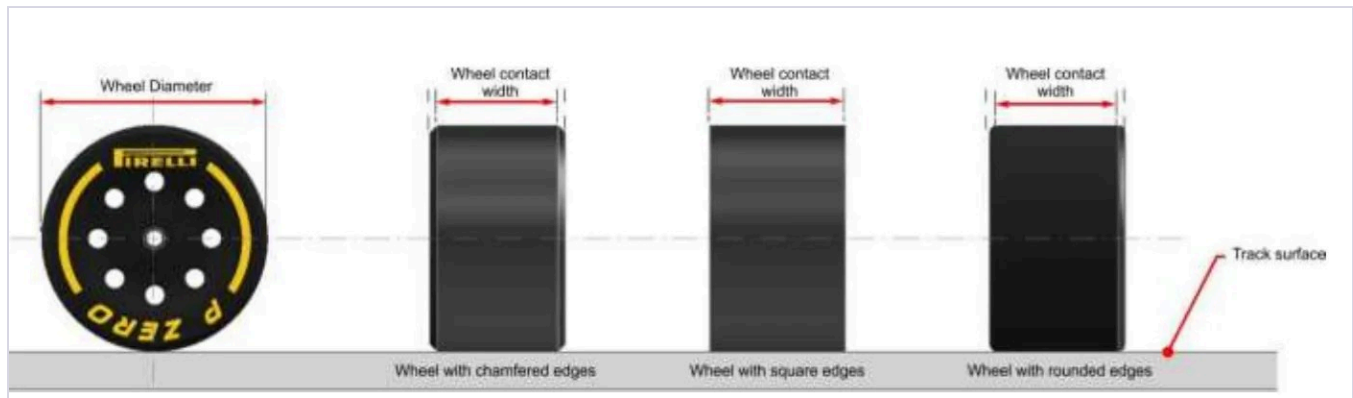
[PERFORMANCE | Penalty – 2.5pts per millimeter per wheel]

This is measured along the surface of the wheel that makes constant contact with the track surface, excluding any chamfers or fillets.

If only one wheel per axle is outside of the minimum or maximum dimension, the point penalty will stand but the regulation is reclassified as General.

T7.4.1 Front wheels - Absolute Min: 13.0mm

T7.4.2 Rear wheels - Absolute Min: 17.0mm



Wheel diameter and track contact width for chamfered, square and rounded wheel edges (T7.4).

T7.5 Diameter

[PERFORMANCE | Penalty – 2.5pts per millimeter per wheel]

This is the wheel diameter measured across the minimum track contact width rolling surface.

If only one wheel per axle is outside of the minimum or maximum dimension, the point penalty will stand but the regulation is reclassified as General.

Absolute Min: 28.0mm / **Absolute Max:** 32.0mm

T7.6 Race track contact

[PERFORMANCE | Penalty – 2.5pts per wheel]

All four (4) wheels must touch the racing surface at the same time across the full track contact width, measured with a full power unit cartridge inserted with the car sitting on all four (4) wheels with no outside assistance. Race track contact must be maintained prior to car launch and during racing.

T7.7 Rolling surface

[PERFORMANCE | Penalty – 2.5pts per wheel]

The wheel diameter must be consistent across the track contact width. (i.e. no tread like features are permitted)

T7.8 Rotation

[PERFORMANCE | Penalty – 5pts per wheel]

To facilitate forward motion of the car during racing all (4) wheels must rotate freely about their axis using a 3° maximum inclined surface.

Absolute Max: 3°

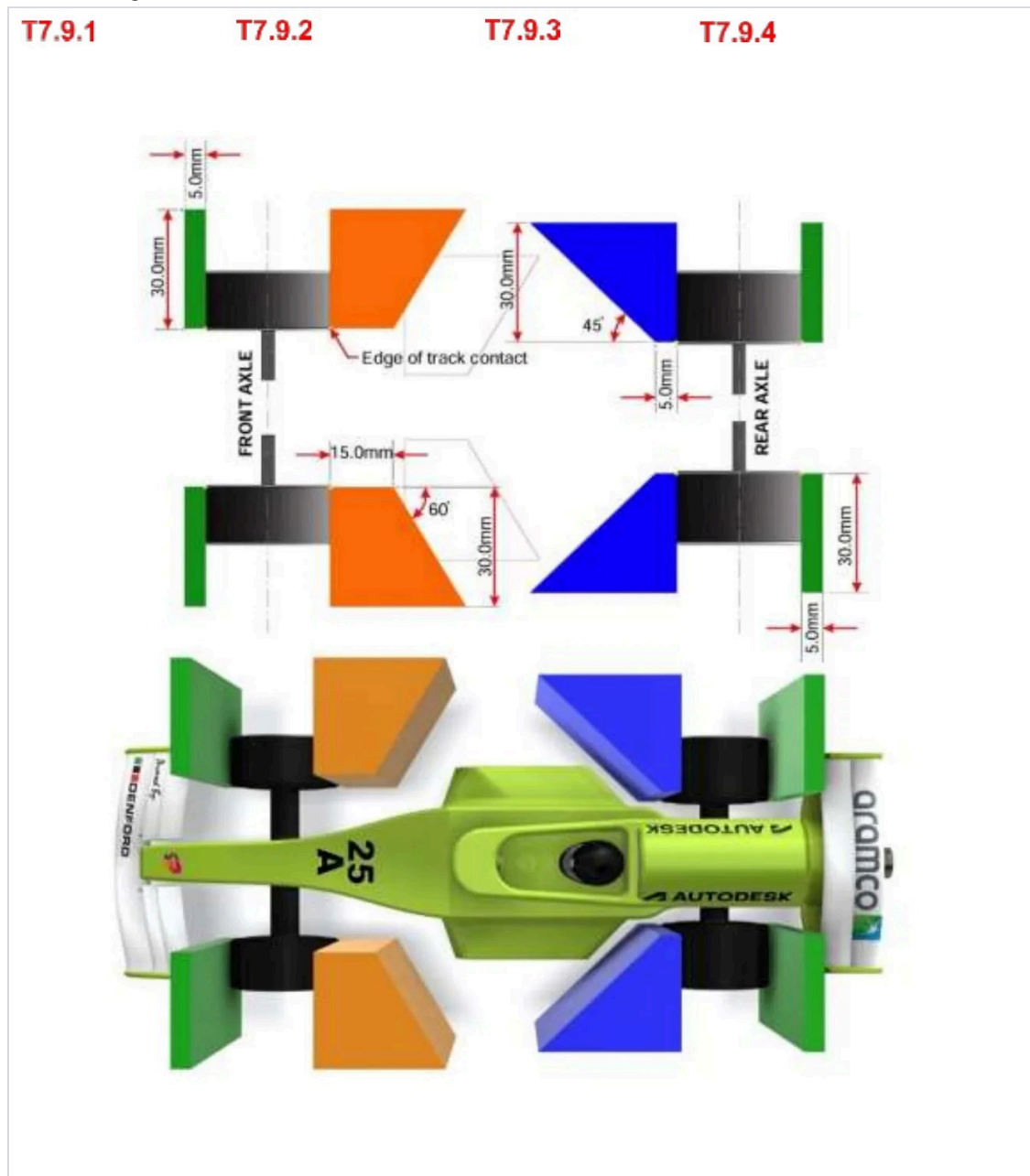
T7.9 Visibility in top and bottom views

[PERFORMANCE]

The visibility of all wheels must not be physically obscured by any component of the car in the car's top and bottom elevation views. Car body or any other components must not exist within the dimensions illustrated below. These dimensions must exist from the inside edges of each wheels' track contact width to the extreme width of the car assembly and a height from track surface of 65.0mm. This is measured, parallel to the vertical reference plane and track surface. Please also refer to T8.6 and T9.6 – Clear Airflow.

REGULATION	LOCATION	DIMENSION	PENALTY
T7.9.1	In front of front wheels	Absolute Min: 5.0mm	2.5pts
T7.9.2	Behind front wheels	Absolute minimum dimensions on diagram below	5pts
T7.9.3	In front of rear wheels	Absolute minimum dimensions on diagram below	5pts
T7.9.4	Behind rear wheels	Absolute Min: 5.0mm	2.5pts

Scrutineering tools used to define these clearances will be manufactured to dimensions as illustrated below.

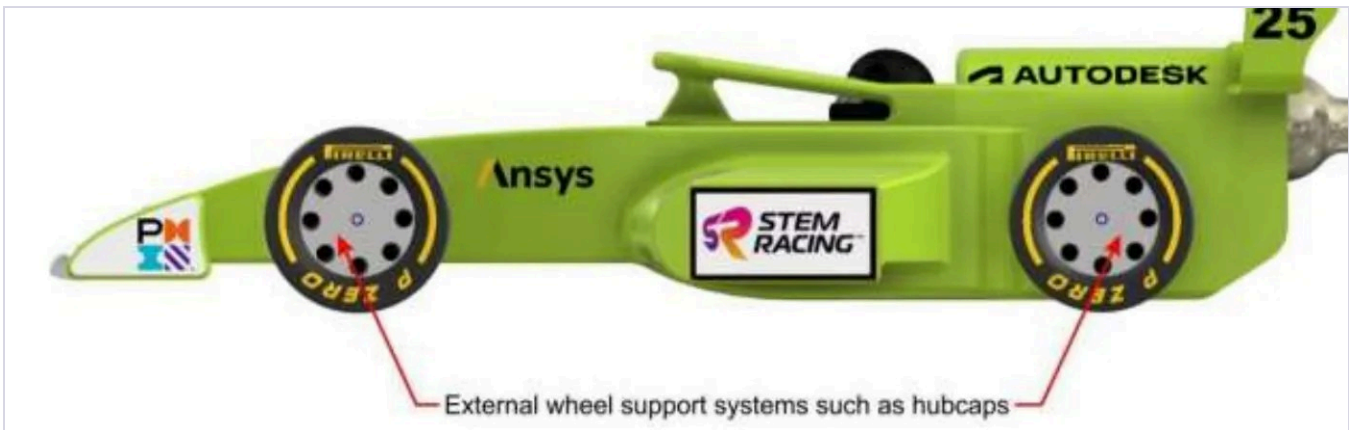


Clearance zones that must be free of car body or other components around the wheels (T7.9.1 to T7.9.4).

T7.10 Visibility in side views

[PERFORMANCE | Penalty – 10pts per wheel]

The visibility of all wheels must not be physically obscured by any component of the car with the exception of any wheel support systems, in the car's side elevation views.



Wheel visibility in the side view (T7.10); external wheel support systems such as hubcaps are allowed.

T7.11 Visibility in front view

[PERFORMANCE | Penalty – 10pts per millimetre]

The visibility of the front wheels in the car's front view may only be physically obstructed to a height of 20.0mm from the track surface. This is measured with a full 8g power unit cartridge inserted into the cartridge chamber with the car sitting on all four (4) wheels with no outside assistance.

Absolute Max: 20mm



Front wheels may be obstructed only up to 20.0mm above the track (T7.11).

T7.12 Wheel support

[GENERAL | Penalty – 5pts each]

T7.12.1 **Wheel support systems:** Wheel support systems may only exist within the cylindrical volume generated through the diameter of the two (2) opposing wheels. Refer to ARTICLE T3.8 Track clearance.



The cylindrical volume generated through the diameter of opposing wheels (T7.12.1).

T7.12.2 **Wheel support systems identification:** The surfaces defining the wheel support structures MUST be dimensioned and identified clearly by hatching, shading or block color within the engineering drawings submitted for scrutineering. (Construction lines are acceptable to define the boundaries between components such as nose cone and wing support structures.) Also refer to the definition in ARTICLE T1.15 for guidance on annotating these features in your Engineering Drawings.

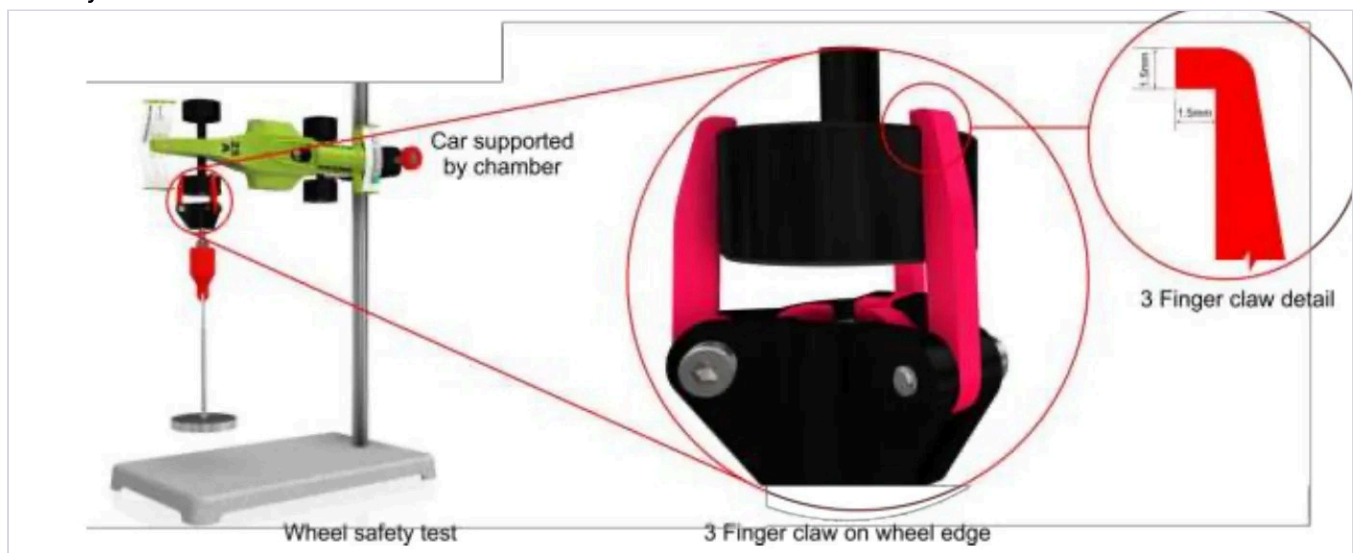
T7.13 Wheel safety test

[SAFETY | Penalty – 2.5pts per wheel]

To prevent wheel detachment during racing and encourage good engineering practice a wheel safety test will be carried out.

The wheels must be robust so as to prevent the diameter or shape changing during racing. The below wheel test will be conducted during scrutineering. A 100g weight will be suspended from each wheel.

To allow for this wheel safety test a minimum clearance (see below) is required from the inner corner of the wheel to the car body.



The wheel safety test, showing the 3 finger claw on the wheel edge and its clearance detail (T7.13).

Article T8 Nose, front wing, end plates and wing

support structures

T8.1 Nose, front wing, end plates and wing support structure identification

[GENERAL | Penalty – 5pts]

The surfaces defining the nose, front wing(s), end plates and wing support structures must be dimensioned and identified clearly by hatching, shading or block color within the engineering drawings submitted for scrutineering. (Clearly labelled construction lines are acceptable to define the boundaries between components such as nose cone and wing support structure). Also refer to the definition in ARTICLE T1.15 for guidance on annotating these features in your Engineering Drawings.

T8.2 Nose cone assembly dimension

[GENERAL | Penalty – 5pts each]

This is measured in front of and parallel to reference plane A to the extreme front of the fully assembled car. From the Reference plane A the nose cone overhang is 40mm maximum.

Absolute Max: 40.0mm



Nose cone overhang, maximum 40.0mm in front of reference plane A (T8.2).

T8.3 Front wing(s) description and placement

[General | Penalty – 5pts]

The design of the car should resemble an actual F1® car through the inclusion of a wing(s) on the nose of the car. Each wing section must have a leading edge and a trailing edge across its full span. Refer to the definition in ARTICLE T1.5.

T8.4 Front wing(s) construction and rigidity

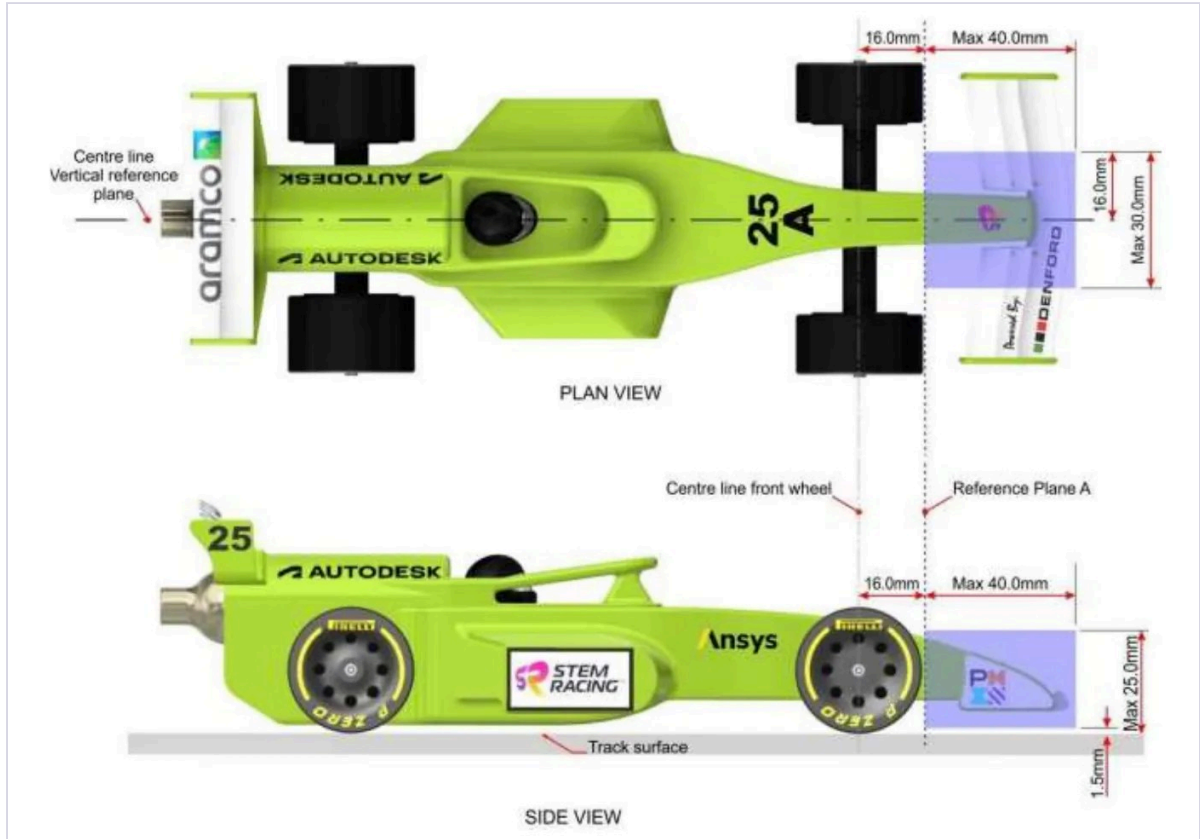
[General | Penalty – 5pts]

The nose, front wing(s) and any support structures may be manufactured from any separate materials. The wing span dimension must remain unchanged during races (i.e. wings must be rigid, ruled at the judge's discretion).

T8.5 Nose, front wing(s), front wing end plates and wing support location

T8.5.1 Nose and wing support structure location - [General | Penalty – 10pts] The nose and front wing support structure must be in front of reference plane A, no more than 25.0mm above the track surface and no wider than 15mm either side of the center line reference plane.

Absolute Max height: 25.0mm

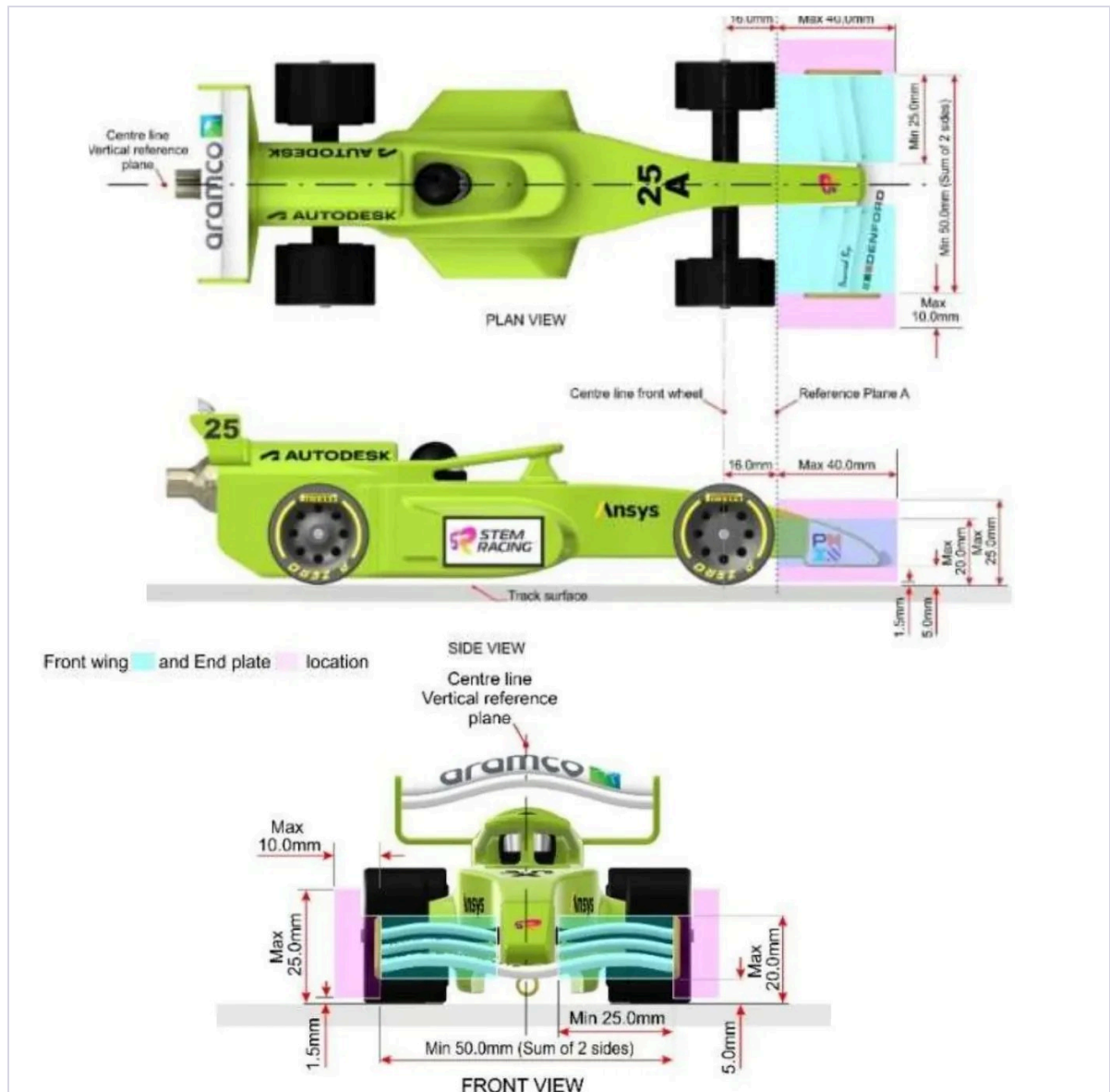


Zone for the nose and front wing support structure (T8.5.1).

T8.5.2

The front wing and front wing end plate location - [General | Penalty – 10pts] The front wing and front wing end plates must be in front of reference plane A. The front wing must be no more than 20.0mm above the track surface when wider than 15mm either side of the center line.

Absolute Max height: 20.0mm



Zones for the front wing and end plates (T8.5.2 and T8.5.3).

T8.5.3

Front wing end plate location - [General | Penalty – 10pts] The front wing endplates must be positioned outside the minimum legal span, with a maximum width of 10.0mm when measured perpendicular to the vertical reference plane, and must not extend more than 25.0mm above the track surface when positioned outside (beyond) the front wheels.

Absolute Max Width: 10.0mm / Absolute Max Height: 25.0mm

The endplates may also play a role in breaking the beam that detects cars crossing the finish line. To maximize accuracy in timing and classification, they should be designed to ensure they reliably trigger the beam as early as possible. Considerations should include their height, width, and position relative to the beam to optimize detection.

Please see Appendix i: Start Box and Finish Gate for beam information.

T8.6 Front wing dimensions

[PERFORMANCE]

T8.6.1 Front wing span - [PERFORMANCE | Penalty – 2pts per millimeter] The front wing span will be measured at right angles to the vertical reference plane along the leading edge, trailing edge, top surface and bottom surface of the wing, whichever is shortest will be considered as the maximum span. Where the front wing span is intersected by another part of the car, the total span is the sum of a maximum of two (2) wing segments, which must be no less than 25.0mm each. (See illustration below)

Absolute Min: 25.0mm x 2 = 50.0mm or 50.0mm x 1 = 50.0mm

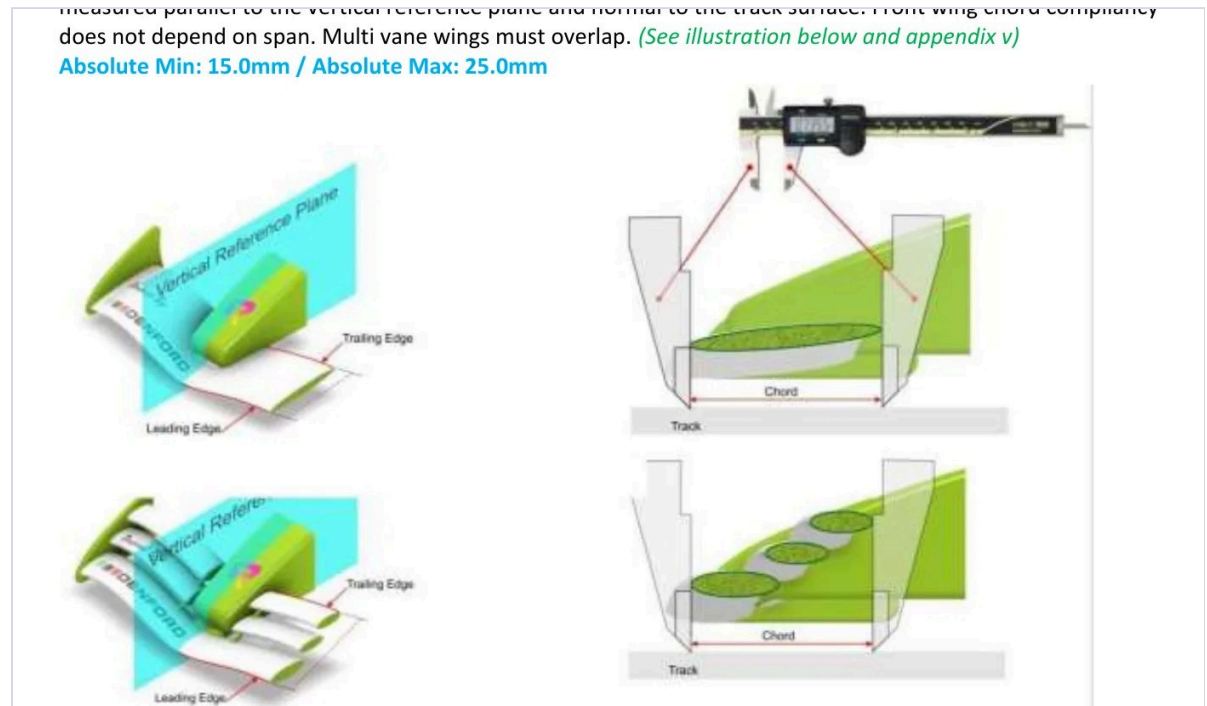


Front wing span, minimum 25.0mm on each side of the intersecting part (T8.6.1).

T8.6.2

Front wing chord - [PERFORMANCE | Penalty – 1pt per millimeter] The front wing chord minimum to maximum dimensions must exist throughout the existing wing span. The chord is the distance between the leading edge and trailing edge of up to three elements that must overlap, it will be measured parallel to the vertical reference plane and normal to the track surface. Front wing chord compliancy does not depend on span. Multi vane wings must overlap. (See illustration below and appendix v)

Absolute Min: 15.0mm / Absolute Max: 25.0mm

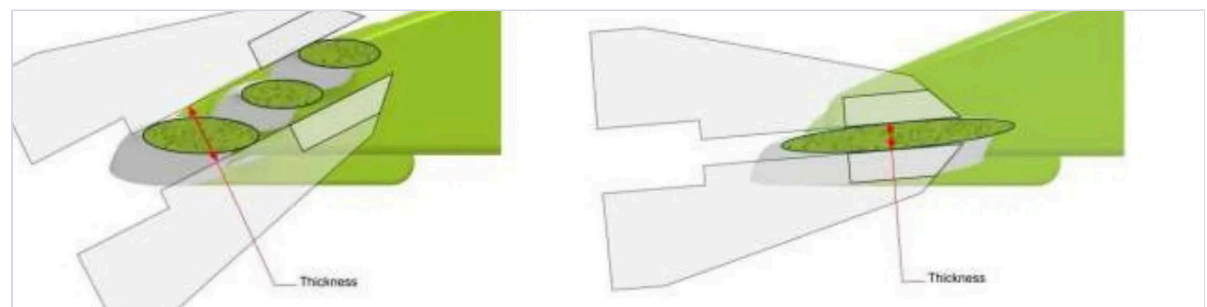


How front wing chord is measured (T8.6.2).

T8.6.3

Front wing thickness - [PERFORMANCE | Penalty – 1pt per millimeter] The wing thickness minimum to maximum dimensions must exist throughout the wing's existing span and at a point along the existing chord. Front wing thickness compliancy does not depend on span and / or chord. (See illustration below)

Absolute Min: 2.0mm / Absolute Max: 6.0mm



How front wing thickness is measured (T8.6.3).

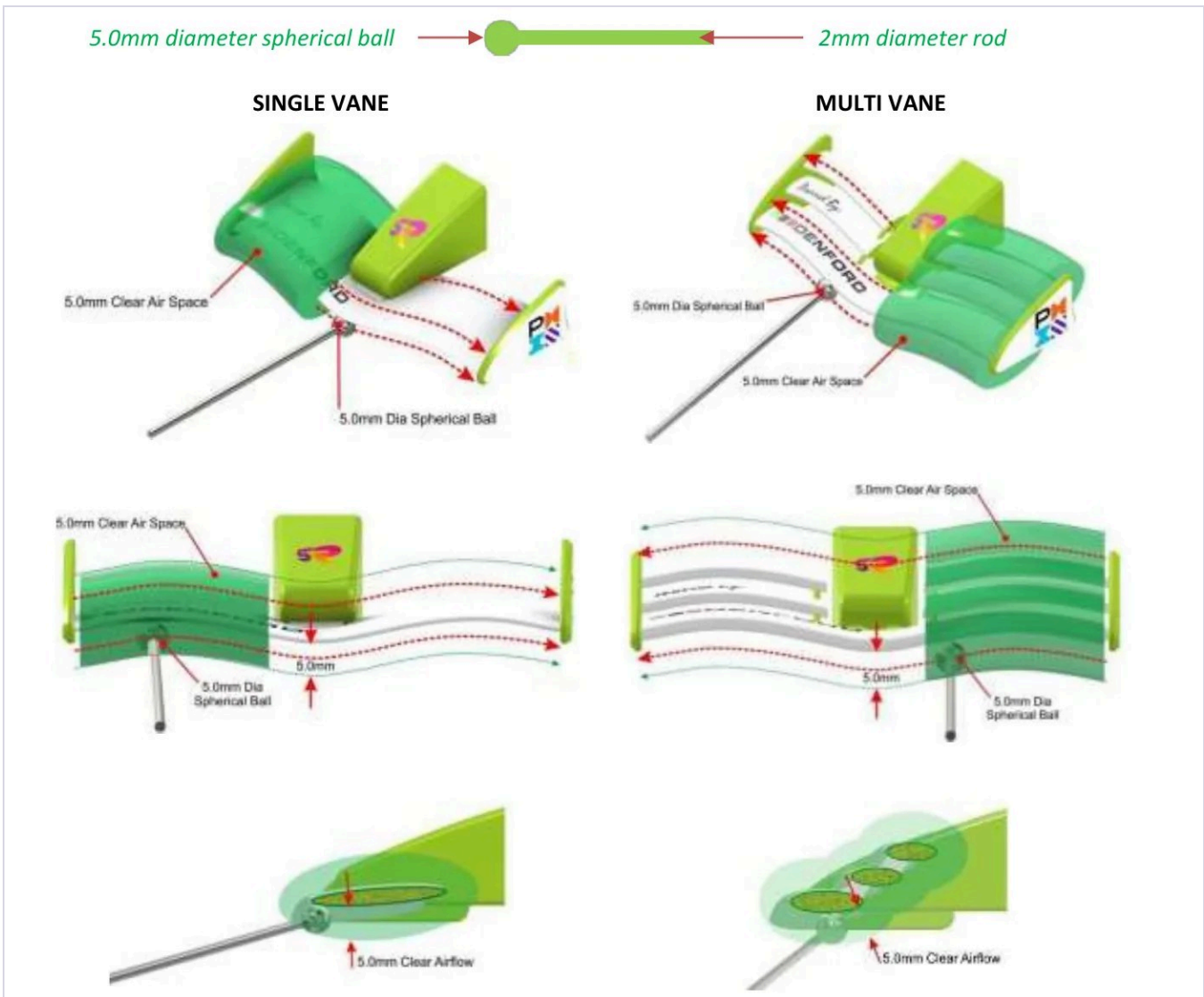
T8.7 Front wing clear airflow

[PERFORMANCE | Penalty – 5pts]

The front wing(s), measured across its / their minimum existing span, must have a minimum of 5.0mm of clear 'air' space to any other part of the car or track surface, measured normal from any part of the wings surface.

Absolute Min: 5.0mm

(To allow for fillet rads where a wing joins a support structure this will be checked with a 5.0mm diameter spherical ball on the end of a 2mm diameter rod.)



Clear air space around the front wing, checked with a 5.0mm ball on a 2mm rod (T8.7).

T8.8 Front wing visibility

[PERFORMANCE | Penalty – 10pts]

Visibility of the front wing(s) must not be physically obstructed by any other component when viewed in the front view.



The front wing must be fully visible in the front view (T8.8).

Article T9 Rear wing and wing support structures

T9.1 Rear wing and wing support structure identification

[General | Penalty – 5pts]

The surfaces defining the rear wing and wing support structure(s) must be dimensioned and identified clearly by hatching, shading or block color within the engineering drawings submitted for scrutineering. (Clearly labeled construction lines are acceptable to define the boundaries between components such as car body and wheel support structures). Also refer to the definition in ARTICLE T1.15 for guidance on annotating these features in your Engineering Drawings.

T9.2 Rear wing description and placement

[General | Penalty – 5pts]

The design of a STEM Racing™ India car should resemble an actual F1® car through the inclusion of a wing at the rear of the car. To be considered a wing section it must have a leading edge and a trailing edge across its required span. Refer to the definition in ARTICLE T1.5

T9.3 Rear wing construction and rigidity

[General | Penalty – 5pts]

The rear wing and any support structures may be manufactured from any separate materials. The wing span dimension must remain unchanged during races (i.e. wings must be rigid, ruled at the judge's discretion).

T9.4 Rear wing, wing support structure and rear overhang location

T9.4.1 Rear wing and wing support structure location [General | Penalty – 10pts] The whole of the rear wing and any support structure must be to the rear of reference plane B.

T9.4.2 Rear overhang length [General | Penalty – 5 pts] This is measured to the rear of and parallel to reference plane B to the extreme rear of the fully assembled car. From the Reference plane B the rear wing and wing support structure overhang is a maximum of 40.0mm.

Absolute Max: 40.0mm

T9.4.3 Rear overhang height [General | Penalty – 5 pts] This is measured to the rear of reference plane B to the highest point of the fully assembled car from the track surface.

Absolute Max: 65.0mm



Rear overhang length and height limits behind reference plane B (T9.4.2 and T9.4.3).

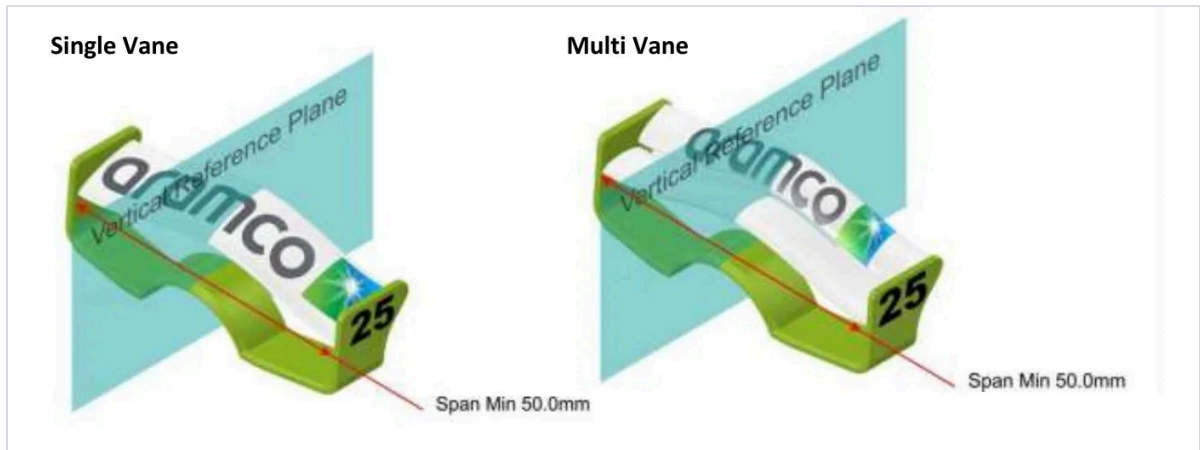
T9.5 Rear wing dimensions

[PERFORMANCE]

T9.5.1

Rear wing span - [PERFORMANCE | Penalty - 2pts per millimeter] The rear wing span will be measured at right angles to the vertical reference plane along the leading edge, trailing edge, top surface or bottom surface of the wing, whichever is shortest will be considered as the maximum span. The rear wing **must** exist as a single, unbroken minimum span of 50.0mm. (See illustration below)

Absolute Min: 50.0mm

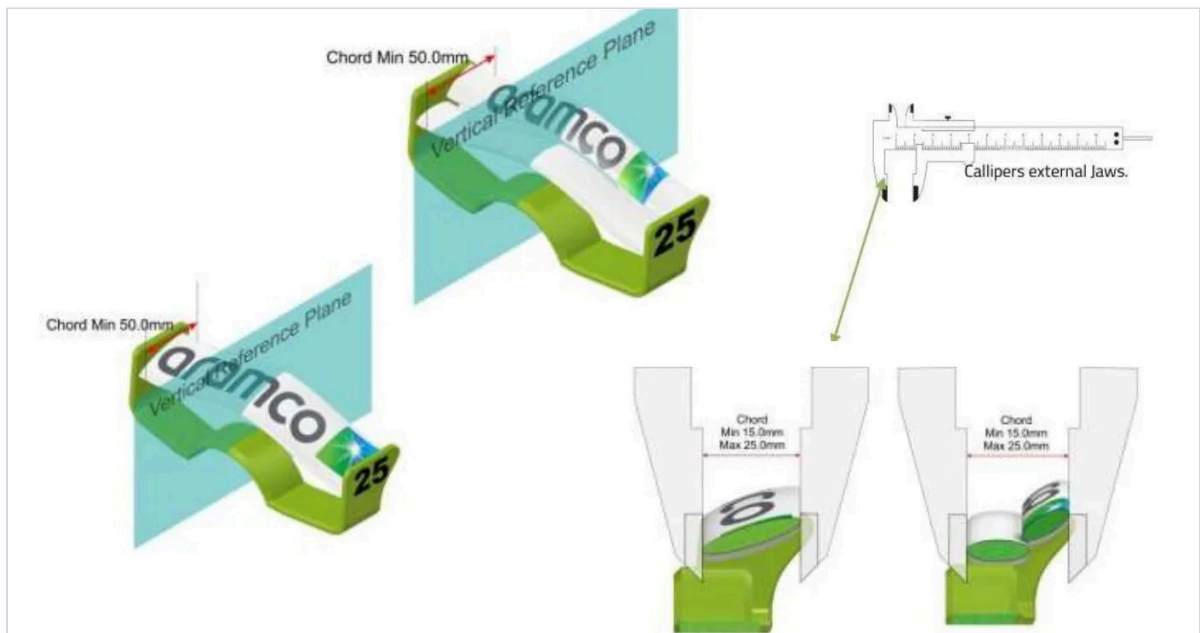


Rear wing span, minimum 50.0mm (T9.5.1).

T9.5.2

Rear wing chord - [PERFORMANCE | Penalty - 1pt per millimeter] The rear wing chord minimum to maximum dimensions must exist throughout its existing wing span. The chord is the distance between the leading edge and trailing edge of up to two elements that must overlap, it will be measured parallel to the vertical reference plane and normal to the track surface. Rear wing chord compliancy does not depend on span. Multi vane wings must overlap. (See illustration below and appendix v)

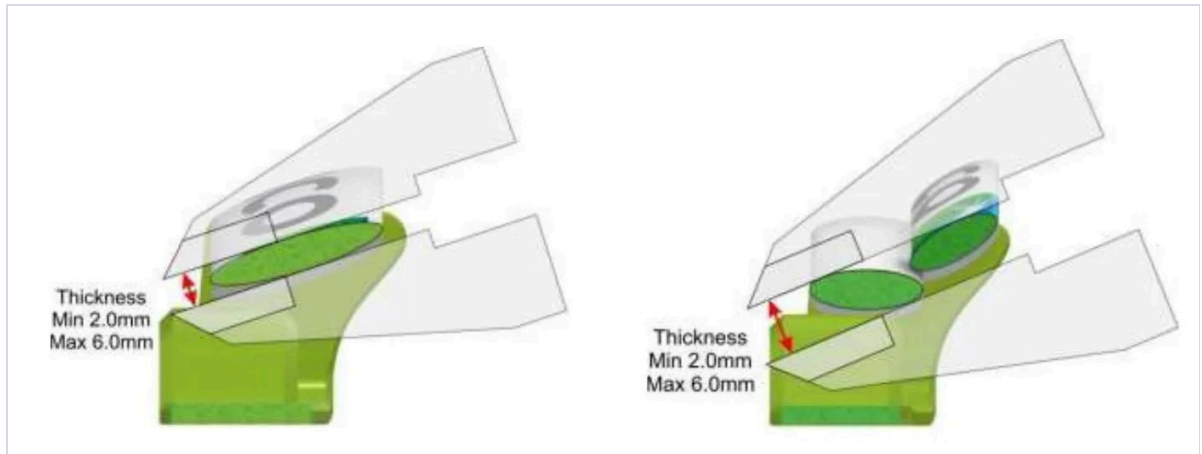
Absolute Min: 15.0mm / Absolute Max: 25.0mm



How rear wing chord is measured with callipers (T9.5.2).

T9.5.3

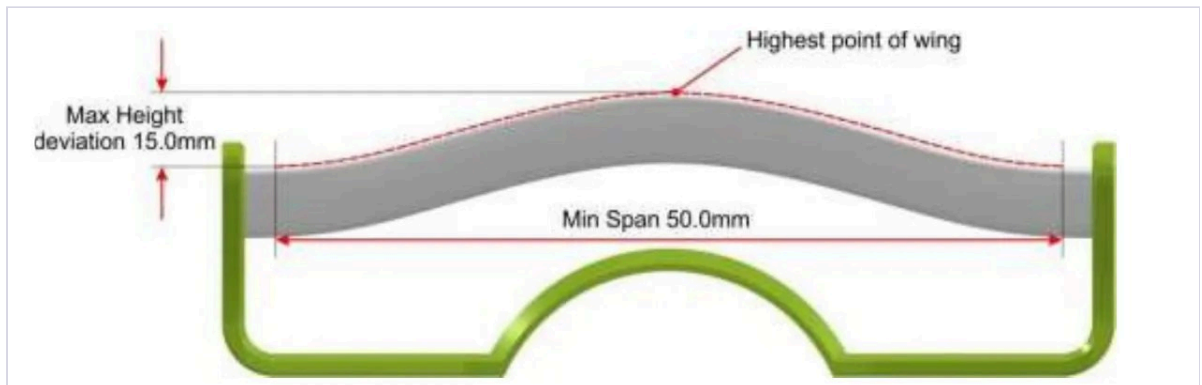
Rear wing thickness - [PERFORMANCE | Penalty - 1pt per millimeter] The wing thickness minimum to maximum dimensions must exist throughout the wing's existing span and at a point along the existing chord. Rear wing thickness compliancy does not depend on span and / or chord. (See illustration below)
Absolute Min: 2.0mm / Absolute Max: 6.0mm



How rear wing thickness is measured (T9.5.3).

T9.5.4

Rear wing height deviation - [GENERAL | Penalty - 1pt per millimeter] The maximum change in height deviation from the highest point of the wing is 15mm. Measured along the minimum span on the top surface.
Absolute maximum height deviation: 15.0mm



Rear wing height deviation, maximum 15.0mm along the minimum span (T9.5.4).

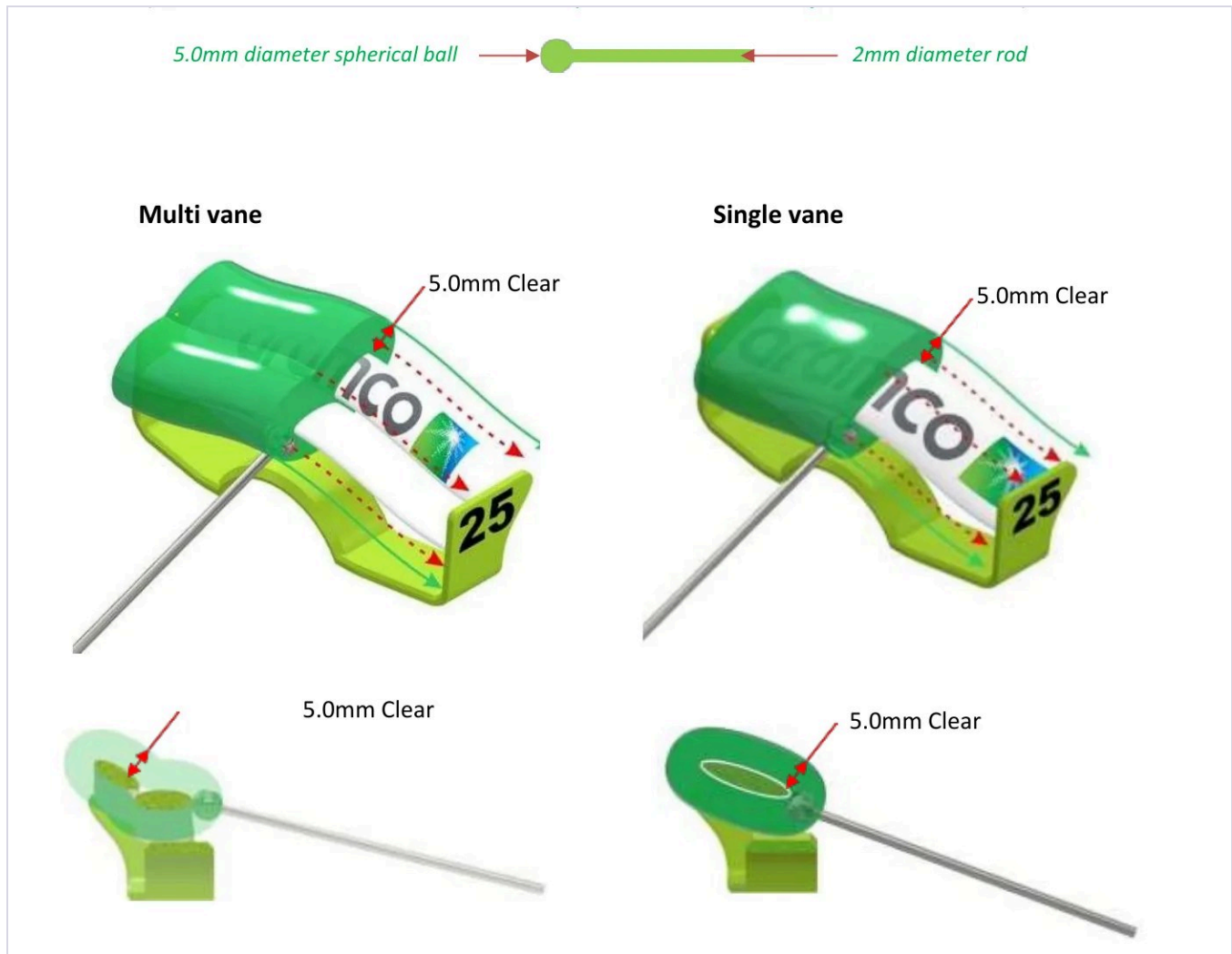
T9.6 Rear wing clear airflow

[PERFORMANCE | Penalty – 5pts]

The rear wing, measured across its existing span, must have a minimum of 5.0mm of clear 'air' space to any other part of the car or track surface, measured normal from any part of the wings surface.

Absolute Min: 5.0mm

(This will be checked with a 5.0mm diameter spherical ball on the end of a 2mm diameter rod.)



Clear air space around the rear wing, checked with a 5.0mm ball on a 2mm rod (T9.6).

T9.7 Rear wing visibility

[PERFORMANCE | Penalty – 10pts]

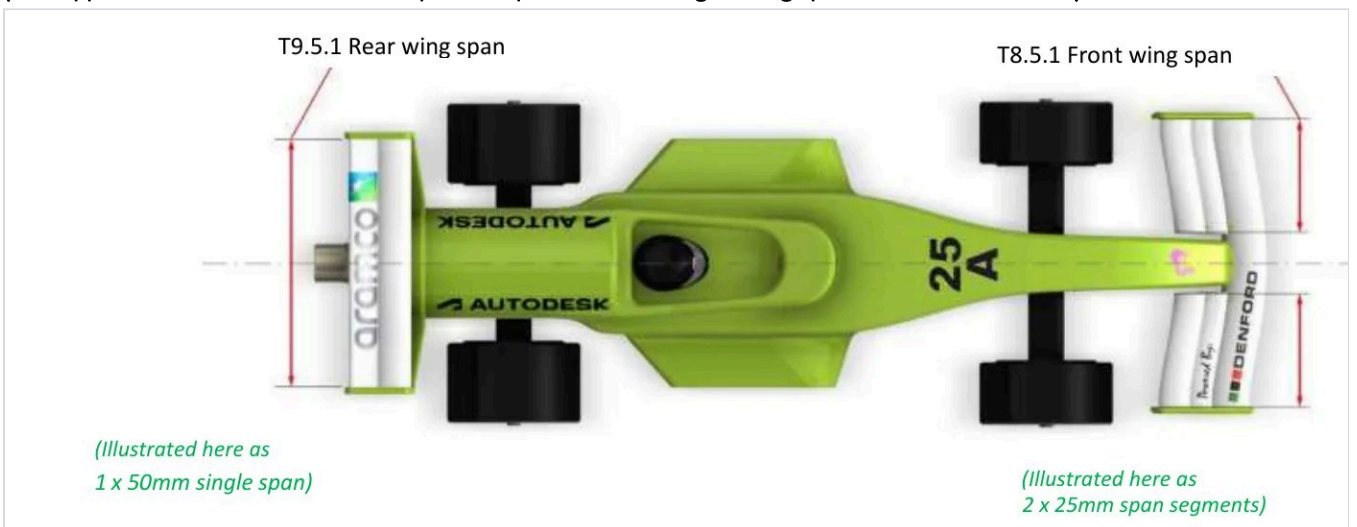
Visibility of the rear wing must not be physically obstructed by any other component when viewed in the front view



The rear wing must be fully visible in the front view (T9.7).

Illustrations for T8 and T9 below:

(See Appendix iii for more detail, examples and penalties relating to wing span, chord and thickness)



Rear wing span as one 50mm span and front wing span as two 25mm segments (T9.5.1 and T8.5.1).

Article T10 Additional components

T10.1 Description and placement

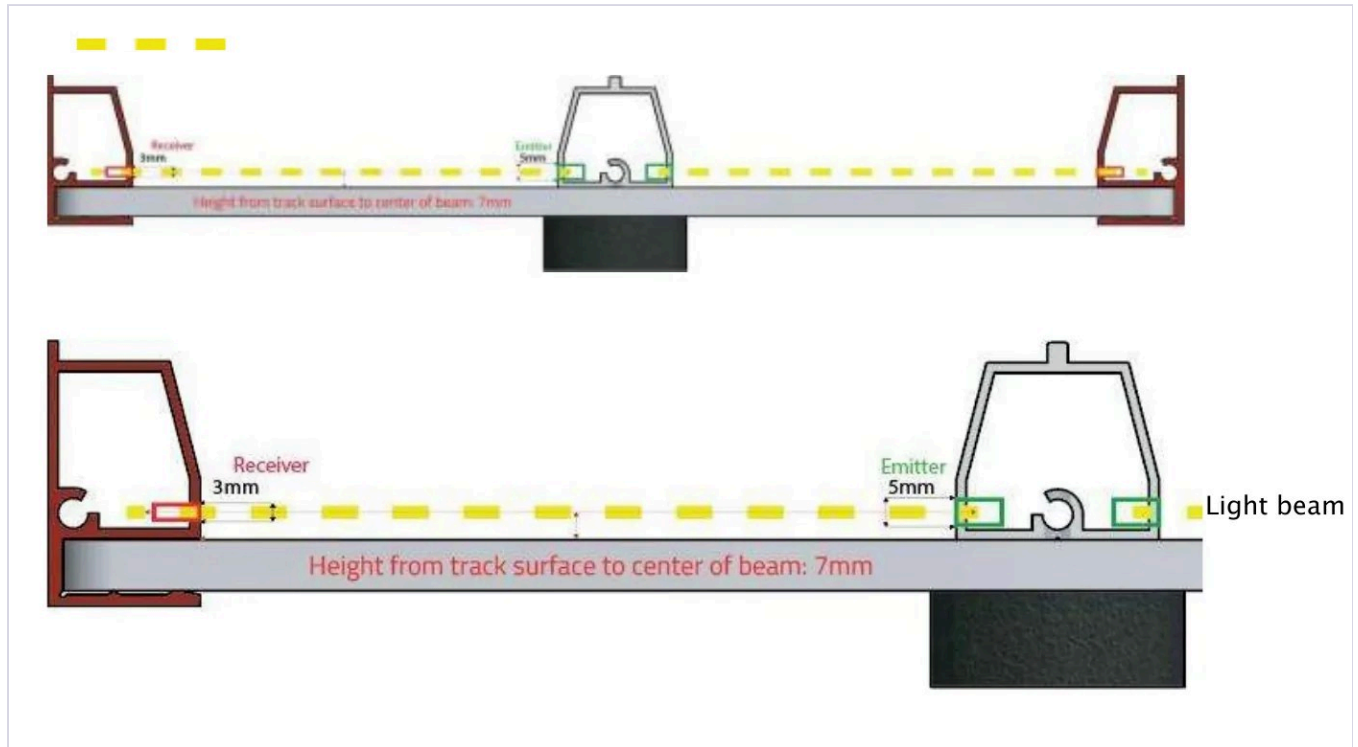
[GENERAL]

Only the power unit cartridge, as positioned by race officials, is permitted to make contact with the launch pods and/or cars prior to and/or during racing. Please refer also to the STEM Racing™ India National Finals 2025/26 Competition Regulations.

Appendix i Start box and finish gate

The start boxes are designed to sit centrally within each lane of the track.

The distance from the emitter center line to the race track surface on both lanes is ~7mm.



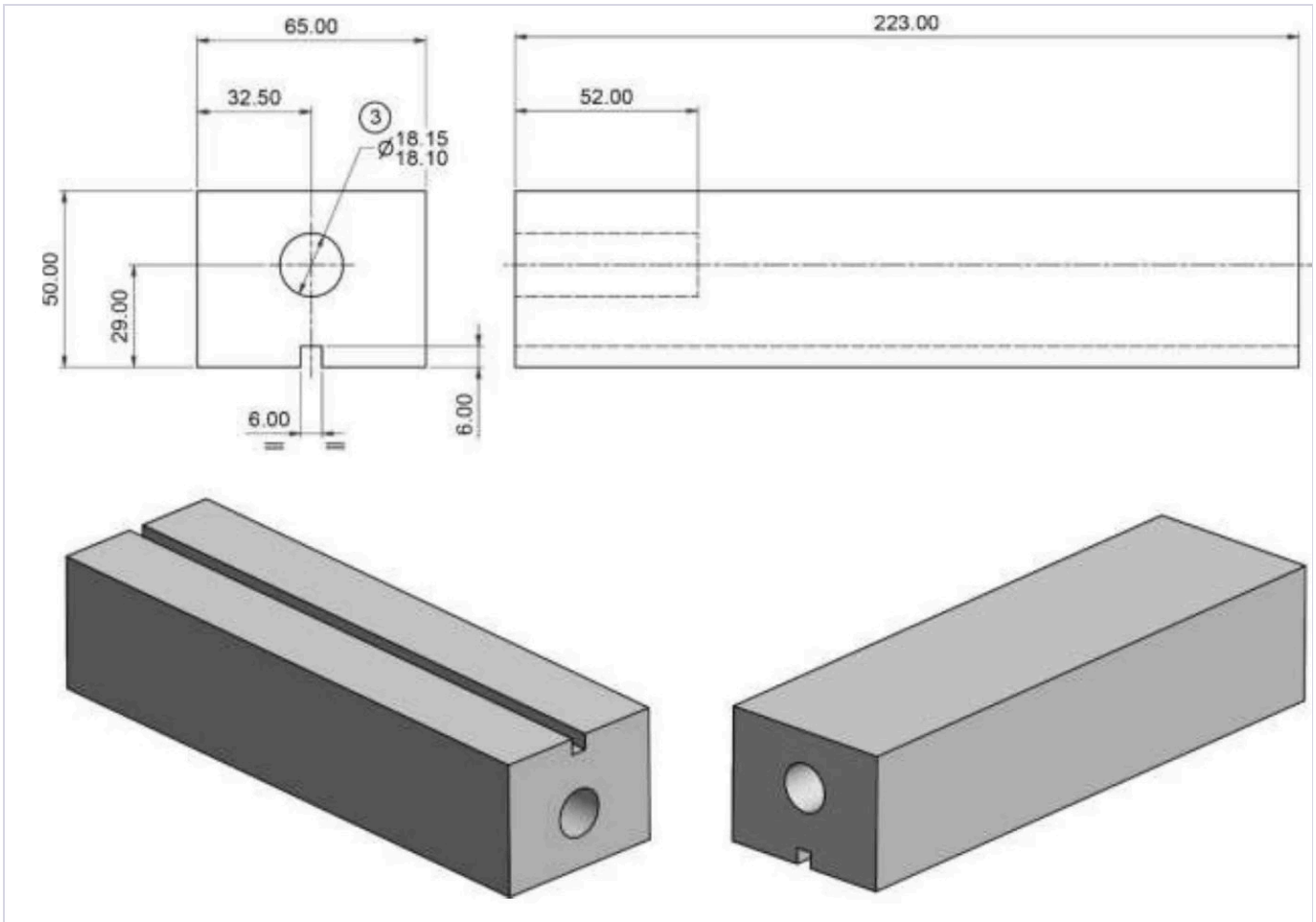
Start box and finish gate: the light beam runs 7mm above the track surface (Appendix i).

Appendix ii Official STEM Racing™ Model Block dimensions

Below: orthographic projection of STEM RACING™ Model Block. All dimensions shown in millimeters

This component is available to download as a universal 3D part from the below link:

https://drive.google.com/file/d/1Pkl5rM2LZHw7w6vUz8_3LgWxHWf94N5x/view?usp=drive_link

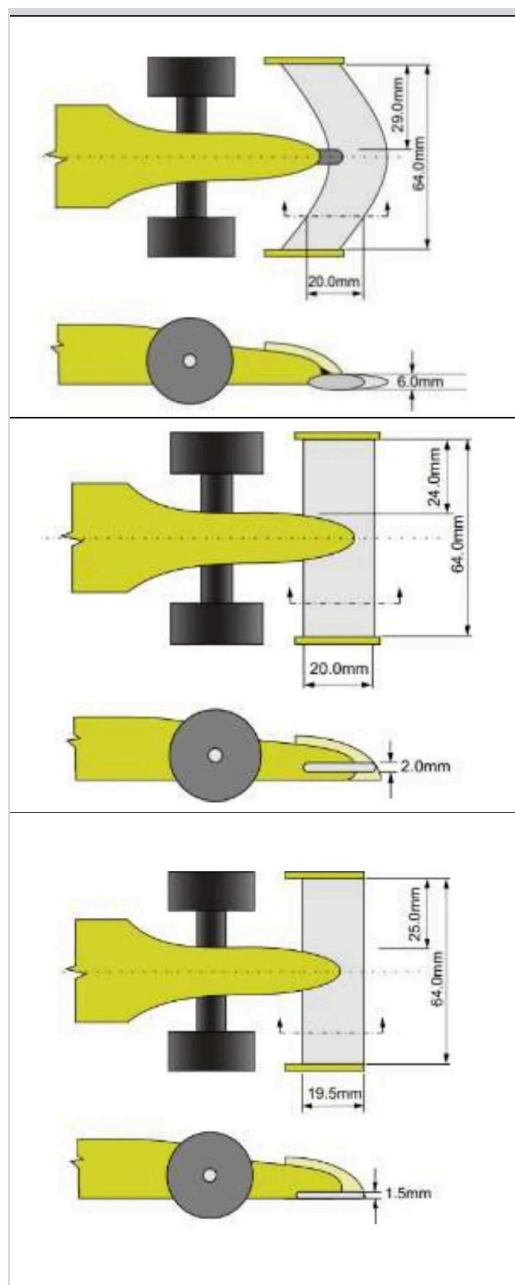


Official model block dimensions in millimeters (Appendix ii).

Appendix iii Front wing compliance examples

The following table shows how penalty points are awarded for front wing span, chord or thickness dimensions that do not meet the specifications set out in T8.5.1, T8.5.2 and T8.5.3.

This is not an exhaustive list of all possible wing configurations.

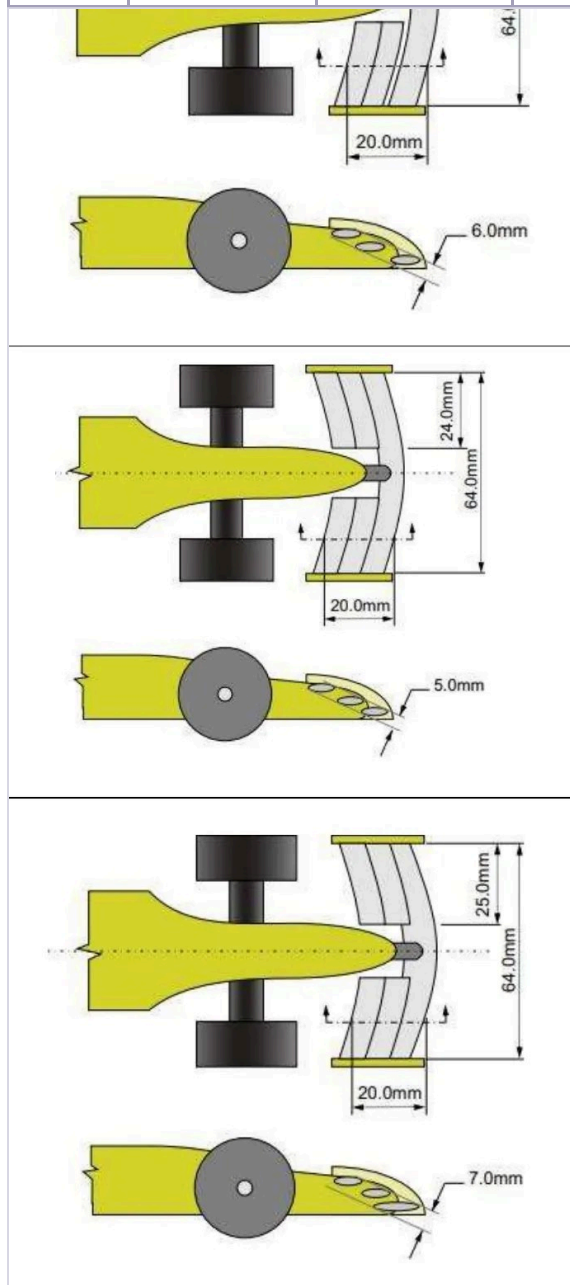


Drawings for the single vane front wing examples 1 to 3, top to bottom (Appendix iii).

Front Wing Examples – Single Vane

EXAMPLE	T8.5.1 SPAN (-2PTS PER MM)	T8.5.2 CHORD (-1PTS PER MM)	T8.5.3 THICKNESS (-1PTS PER MM)	REMARKS
Example 1	PASS	PASS	PASS	Wing span is split by single mounted central wing support. Wing span segments are minimum length, therefore PASS Wing chord is minimum throughout the existing span, therefore PASS. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS.
Example 2	FAIL	PASS	PASS	Wing span is split by nose. Wing span segments are less than minimum length at trailing edge (rear of wing), therefore FAIL Wing chord is minimum throughout the existing span, therefore PASS. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS.

EXAMPLE	T8.5.1 SPAN (-2PTS PER MM)	T8.5.2 CHORD (-1PTS PER MM)	T8.5.3 THICKNESS (-1PTS PER MM)	REMARKS
Example 3	PASS	PASS	FAIL	Wing span is split by nose. Wing span segments are minimum length, therefore PASS Wing chord is minimum throughout the existing span, therefore PASS. Wing thickness is less than minimum, therefore FAIL.



Drawings for the multiple vane front wing examples 1 to 3, top to bottom (Appendix iii).

Front Wing Examples – Multiple Vanes

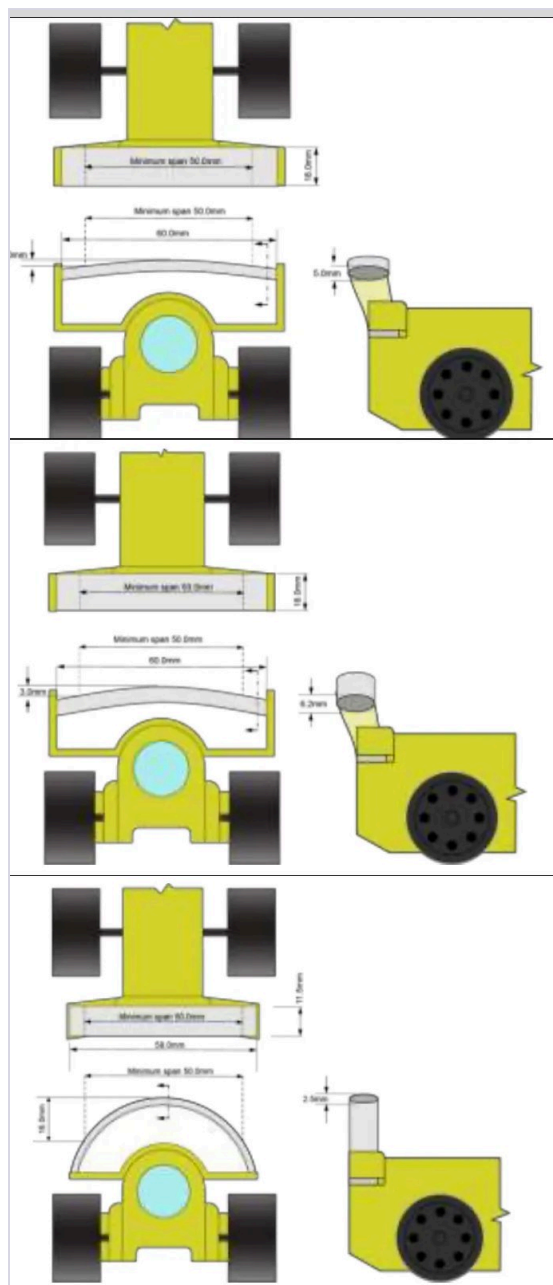
EXAMPLE	T8.5.1 SPAN (-2PTS PER MM)	T8.5.2 CHORD (-1PTS PER MM)	T8.5.3 THICKNESS (-1PTS PER MM)	REMARKS
---------	-------------------------------	--------------------------------	------------------------------------	---------

EXAMPLE	T8.5.1 SPAN (-2PTS PER MM)	T8.5.2 CHORD (-1PTS PER MM)	T8.5.3 THICKNESS (-1PTS PER MM)	REMARKS
Example 1	PASS	FAIL	PASS	Wing span is split by single mounted central wing support. Wing span segments are minimum length (25mm+25mm), therefore PASS Wing chord is minimum throughout the existing span, but Multi vane wings do not overlap FAIL. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS.
Example 2	FAIL	PASS	PASS	Wing span is split by single mounted central wing support. Wing span segments are less than minimum length at trailing edge (rear of wing), therefore FAIL Wing chord is minimum throughout the existing span, therefore PASS. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS.
Example 3	PASS	PASS	FAIL	Wing span is split by single mounted central wing support. Wing span segments are minimum length (25mm+25mm), therefore PASS Wing chord is minimum throughout the existing span, therefore PASS. Wing thickness is over the maximum limit, therefore FAIL.

Appendix iv *Rear wing compliance examples*

The following table shows how penalty points are awarded for rear wing span, chord or thickness dimensions that do not meet the specifications set out in T9.5.1, T9.5.2, T9.5.3, T9.5.4:

This is not an exhaustive list of all possible wing configurations.

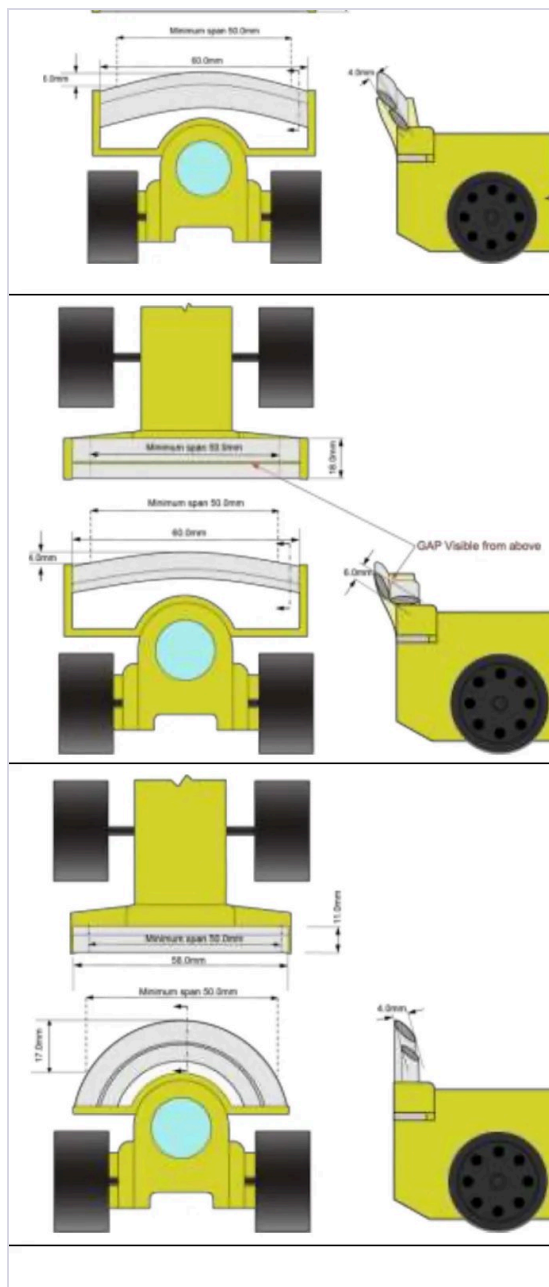


Drawings for the single vane rear wing examples 1 to 3, top to bottom (Appendix iv).

Rear Wing Examples – Single Vane

EXAMPLE	T9.5.1 SPAN (-2PTS PER MM)	T9.5.2 CHORD (-1PT PER MM)	T9.5.3 THICKNESS (-1PT PER MM)	T9.5.4 DEVIATION (-1PT PER MM)	REMARKS
Example 1	PASS	PASS	PASS	PASS	Wing span is minimum length, therefore PASS. Wing chord is minimum throughout the existing span, therefore PASS. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS. Rear wing deviation is below maximum, therefore PASS.

EXAMPLE	T9.5.1 SPAN (-2PTS PER MM)	T9.5.2 CHORD (-1PT PER MM)	T9.5.3 THICKNESS (-1PT PER MM)	T9.5.4 DEVIATION (-1PT PER MM)	REMARKS
Example 2	PASS	PASS	FAIL	PASS	Wing span is minimum length, therefore PASS. Wing chord is minimum throughout the existing span, therefore PASS. Wing thickness is more than maximum, therefore FAIL. Rear wing deviation is below maximum, therefore PASS.
Example 3	PASS	FAIL	PASS	FAIL	Wing span is minimum length, therefore PASS. Wing chord is less than minimum length, therefore FAIL. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS. Wing deviation exceeds maximum height, therefore FAIL.



Drawings for the multiple vane rear wing examples 1 to 3, top to bottom (Appendix iv).

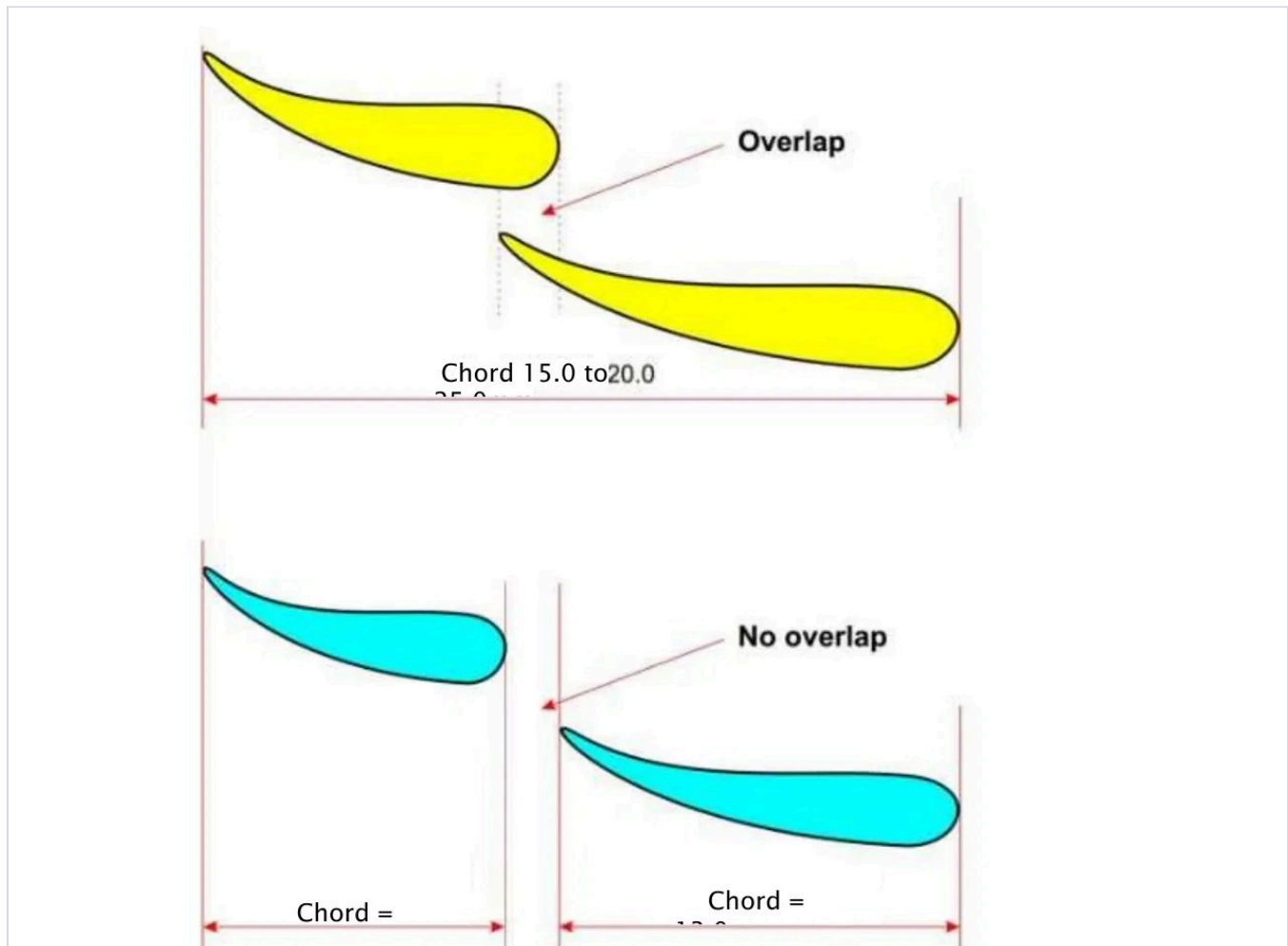
Rear Wing Examples – Multiple Vane

EXAMPLE	T9.5.1 SPAN (-2PTS PER MM)	T9.5.2 CHORD (-1PT PER MM)	T9.5.3 THICKNESS (-1PT PER MM)	T9.5.4 DEVIATION (-1PT PER MM)	REMARKS
Example 1	PASS	PASS	PASS	PASS	Wing span is minimum length, therefore PASS. Wing chord is minimum throughout the existing span, therefore PASS. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS. Wing deviation is minimum throughout the existing span, therefore PASS.

EXAMPLE	T9.5.1 SPAN (-2PTS PER MM)	T9.5.2 CHORD (-1PT PER MM)	T9.5.3 THICKNESS (-1PT PER MM)	T9.5.4 DEVIATION (-1PT PER MM)	REMARKS
Example 2	PASS	FAIL	PASS	PASS	Wing span is minimum length, therefore PASS. Wing chord is minimum throughout the existing span, but Multi vane wings do not overlap FAIL. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS. Wing deviation is minimum throughout the existing span, therefore PASS.
Example 3	PASS	FAIL	PASS	FAIL	Wing span is minimum length, therefore PASS. Wing chord is less than minimum length, therefore FAIL. Wing thickness is minimum throughout the wing's existing span and at a point along the existing chord, therefore PASS. Wing deviation exceeds maximum height, therefore FAIL.

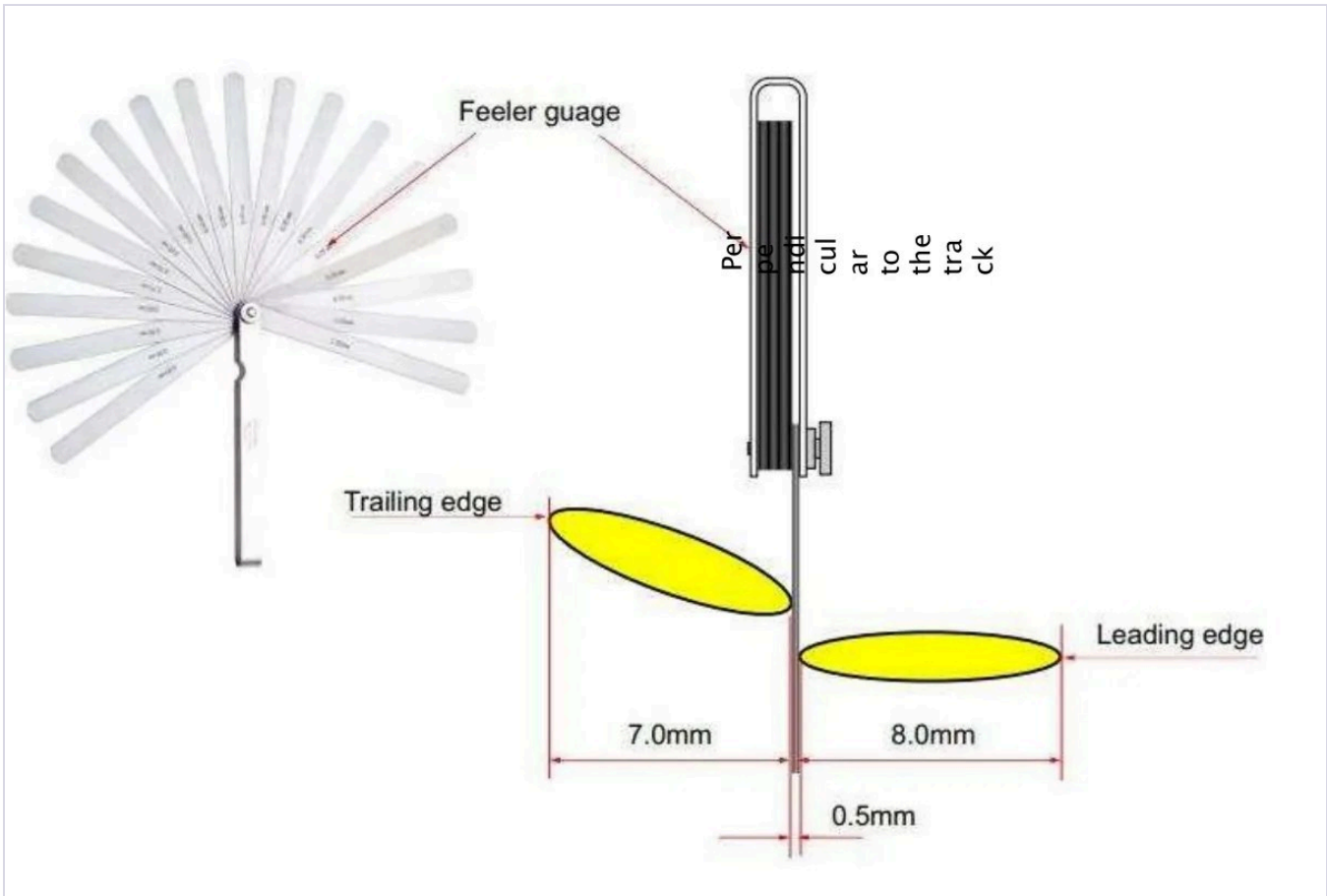
Appendix v Wing overlap

Wing overlap explanation:



Overlapping vanes compared with vanes that do not overlap (Appendix v). Some chord labels are partly clipped in the original drawing.

How to check if a gap exists between the elements:

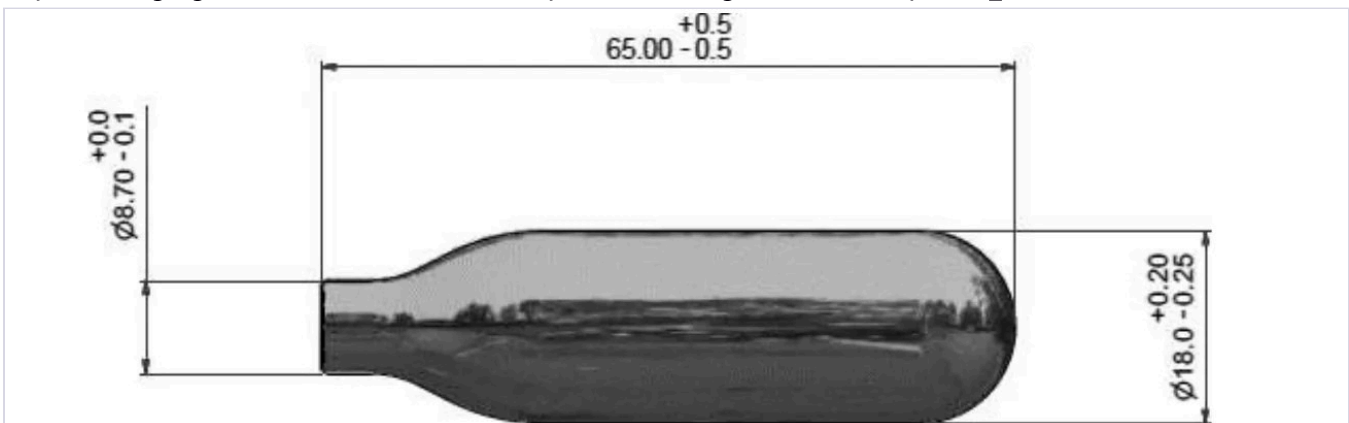


Checking for a gap between the elements with a feeler gauge held perpendicular to the track (Appendix v).

Appendix vi Official Denford Race Power Pack dimensions

This component is available to download as a universal 3D part from the below link:

https://drive.google.com/file/d/1KbsfMYK1KcdzqSJ2Ba7iLrYNOugVN-VH/view?usp=drive_link

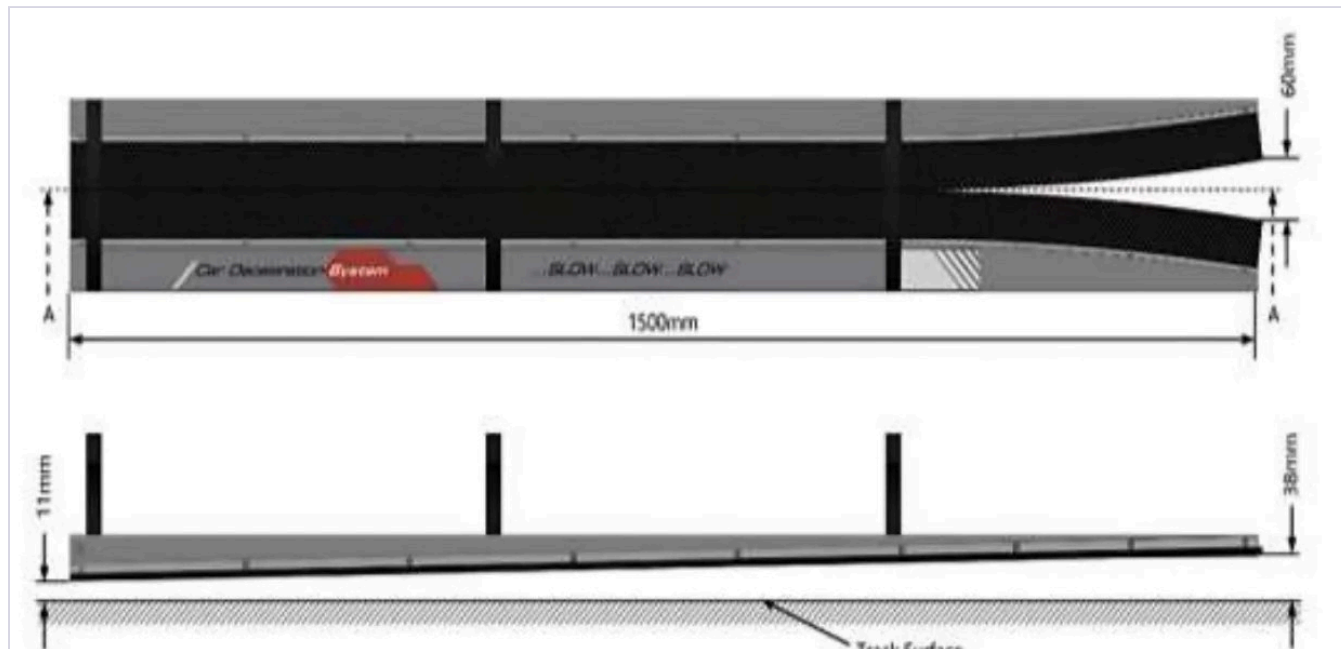


Official Denford Race Power Pack dimensions in millimeters (Appendix vi).

Full weight of RACE Power Pack: between: 28.9g and 29.4g

Appendix vii Car deceleration system

The Deceleration System acts to bring cars to rest once crossing the finish line. STEM Racing™ India will provide Deceleration System which is integrated into the final track section after the finish line.



The car deceleration system after the finish line (Appendix vii).

Please note:

The official STEM Racing™ India deceleration system will be the only permitted stopping system. Please refer to the STEM Racing™ India National Finals 2025/26 Competition Regulations for full details.

Appendix viii Example of component identification

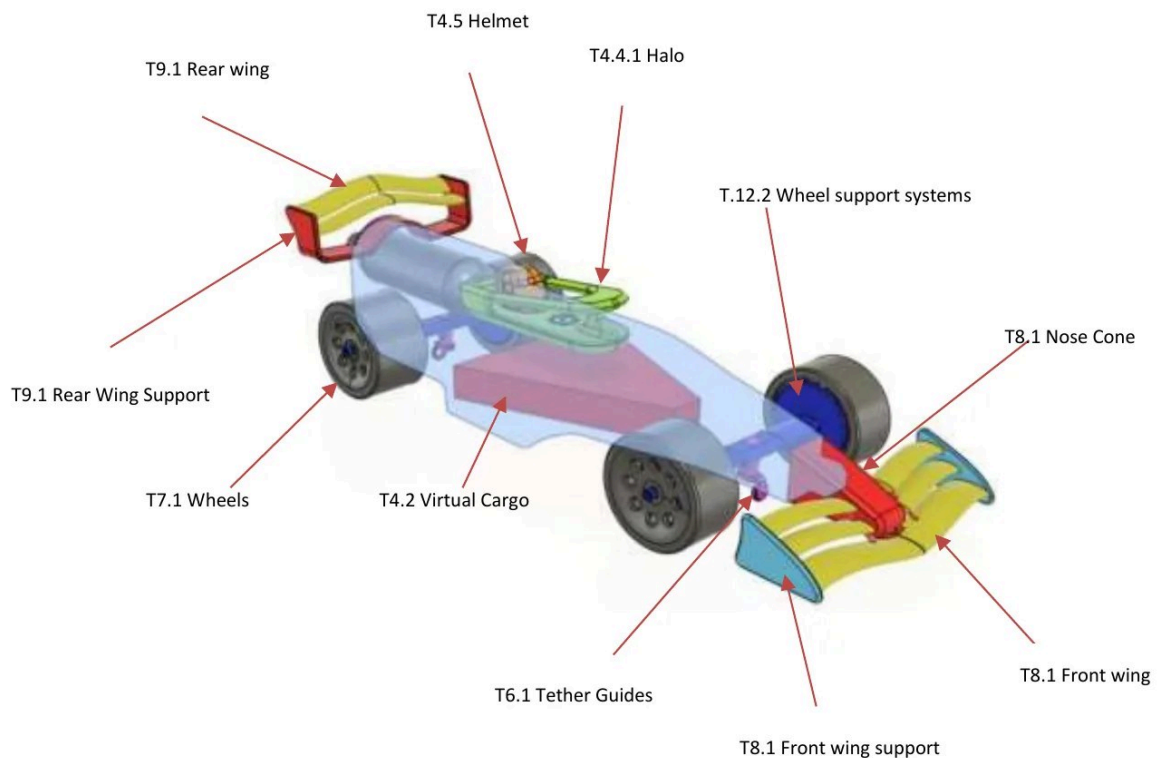
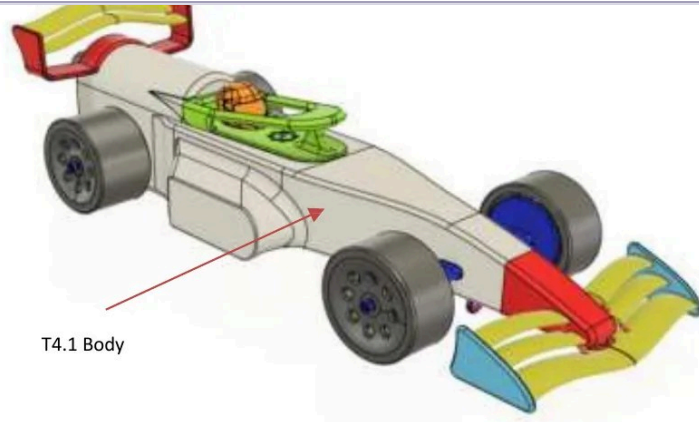
This is a basic example of component identification. Ensure that the regulation numbers align with the 2025/26 Technical Regulations. For detailed guidance on identifying your components, refer to T1.15.

Colour key

COLOUR	COMPONENT
Yellow	Front / Rear Wings
Coral red	Rear Wing Support
Light blue	Front Wing end plate
Red	Nose Cone
Blue	Wheel Support System
Green	Halo
Orange	Helmet
Light grey	Body

COLOUR	COMPONENT
Dark grey	Wheels
Pink	Tether Guides

	Front / Rear Wings
	Rear Wing Support
	Front Wing end plate
	Nose Cone
	Wheel Support System
	Halo
	Helmet
	Body
	Wheels
	Tether Guides



Example of identifying components in an engineering drawing (Appendix viii). One label is printed as T.12.2 in the original.

Appendix ix Halo and ballast container drawings

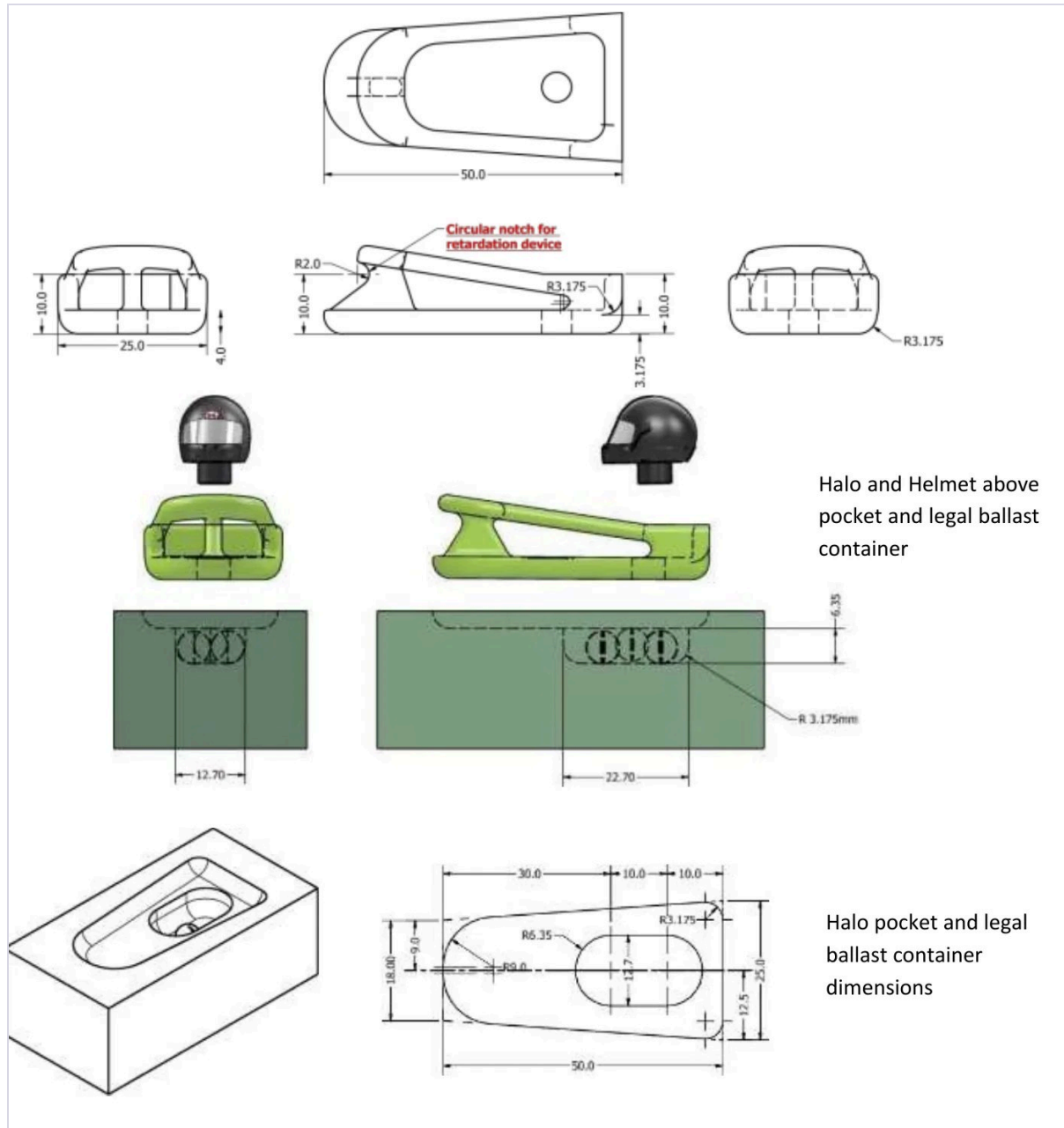
To accommodate the 'Halo', your car will require a pocket cut (which will include a Ballast container directly below the helmet aperture) the dimensions in the drawing below using a 6.35mm (1/4") ball nose tool to a depth of 3.175mm.

The center of the Halo 'Circular notch' for the retardation device is exactly 10.0 mm above the bottom of the pocket. To be effective the center of the Circular Notch must be 34.0mm $\pm$ 1.0mm above the track surface. To achieve this the bottom of the pocket must be 24.0mm above the track in the assembled car. Use the dimensions on the diagrams below to check you

have manufactured your Halo correctly

Legal Ballast

To help achieve the minimum weight, legal ballast may now be added to the legal ballast container under the halo, accessible through the halo/helmet aperture. The Virtual Cargo can coincide with the legal ballast container but not the halo pocket. The helmet should not be glued in place to allow for ballast adjustments before scrutineering.



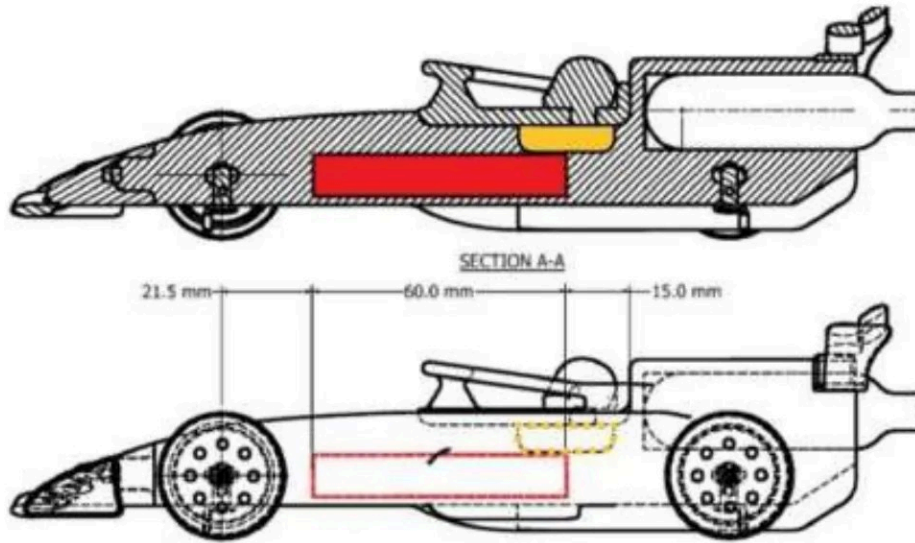
Halo, helmet, pocket and legal ballast container dimensions in millimeters (Appendix ix).



Halo with 6.0mm hole, helmet with 6.0mm spigot and isometric view of the halo (Appendix ix).

Demonstration of compliance of Virtual Cargo and Legal Ballast Container

	Legal ballast Container
	Virtual cargo



The virtual cargo (red) and the legal ballast container (orange) may coincide but not the halo pocket (Appendix ix).

Please Remember:

The Virtual Cargo can coincide with the legal ballast container but not the halo pocket

Clarifications

Please ensure that once you receive the STEM Racing™ India National Finals 2025/26 Competition Regulations, you read through them carefully.

For Official Technical & Competition Regulations Clarifications please email us at competitions@stemplify.com

Work hard, see you at the National Finals!